Mathematics

1,000 Digital SAT Math Questions

Practice items matched to the published Digital SAT math blueprint.

Algebra	350 questions, about 35%
Advanced Math	350 questions, about 35%
Problem-Solving and Data Analysis	150 questions, about 15%
Geometry and Trigonometry	150 questions, about 15%

Of these 1,000 items, 801 are four-option multiple choice and 199 are student-produced responses, the kind you type as a number. On the Digital SAT itself, about 75% are multiple choice and about 25% are student-produced responses. Each item is marked Easy, Medium, or Hard so the bank covers the range from an easier module through a harder second module.

These are original practice items, not College Board questions, and this is not an official SAT test. A calculator is appropriate throughout, as on the Digital SAT. For student-produced responses, enter an integer or a fraction in lowest terms. Do not use a mixed number. Geometry items use the standard reference facts: triangle angle sum 180°, a circle has 360°, $c = 2\pi r$, $A = \pi r^2$, rectangle area ℓw, triangle area $\frac{1}{2}bh$, Pythagorean theorem, 30°-60°-90° and 45°-45°-90° side ratios, cylinder volume $\pi r^2 h$, and right-triangle trigonometry.

The answer key begins after question 1000.

Algebra

1. Linear equations in one variable · Easy · MC
What is the solution to the equation $5x + 11 = 51$?
A. 8
B. 9
C. 40
D. 7

2. Linear equations in one variable · Easy · MC
What is the solution to the equation $2x + 10 = 14$?
A. 2
B. 3
C. 1
D. 4

3. Linear equations in one variable · Easy · SPR
What is the solution to the equation $2x + 8 = 12$?

4. Linear equations in one variable · Easy · MC
What is the solution to the equation $3x + 9 = 42$?
A. 33
B. 10
C. 12
D. 11

5. Linear equations in one variable · Easy · SPR
What is the solution to the equation $2x + 15 = 31$?

6. Linear equations in one variable · Easy · MC
What is the solution to the equation $3x + 9 = 15$?
A. 3
B. 2
C. 1
D. 6

7. Linear equations in one variable · Easy · MC
What is the solution to the equation $6x + 3 = 39$?
A. 5
B. 36
C. 7
D. 6

8. Linear equations in one variable · Easy · MC
What is the solution to the equation $5x + 11 = 71$?
A. 13
B. 12
C. 60
D. 11

9. Linear equations in one variable · Easy · MC
What is the solution to the equation $6x + 15 = 57$?
A. 7
B. 8
C. 42
D. 6

10. Linear equations in one variable · Easy · MC
What is the solution to the equation $4x + 2 = 18$?
A. 4
B. 5
C. 16
D. 3

11. Linear equations in one variable · Easy · MC
What is the solution to the equation $5x + 14 = 59$?
A. 45
B. 9
C. 10
D. 8

12. Linear equations in one variable · Easy · SPR
What is the solution to the equation $6x + 5 = 41$?

13. Linear equations in one variable · Easy · MC
What is the solution to the equation $2x + 3 = 15$?
A. 5
B. 6
C. 7
D. 12

14. Linear equations in one variable · Easy · MC
What is the solution to the equation $2x + 2 = 14$?
A. 7
B. 12
C. 6
D. 5

15. Linear equations in one variable · Easy · SPR
What is the solution to the equation $8x + 9 = 49$?

16. Linear equations in one variable · Easy · MC
What is the solution to the equation $8x + 7 = 31$?
A. 3
B. 4
C. 2
D. 24

17. Linear equations in one variable · Easy · SPR
What is the solution to the equation $3x + 6 = 24$?

18. Linear equations in one variable · Easy · MC
What is the solution to the equation $6x + 6 = 78$?
A. 13
B. 72
C. 12
D. 11

19. Linear equations in one variable · Easy · MC
What is the solution to the equation $4x + 15 = 27$?
A. 2
B. 3
C. 12
D. 4

20. Linear equations in one variable · Easy · SPR
What is the solution to the equation $5x + 2 = 17$?

21. Linear equations in one variable · Easy · SPR
What is the solution to the equation $4x + 13 = 53$?

22. Linear equations in one variable · Easy · MC
What is the solution to the equation $5x + 11 = 56$?
A. 10
B. 9
C. 45
D. 8

23. Linear equations in one variable · Easy · SPR
What is the solution to the equation $5x + 8 = 23$?

24. Linear equations in one variable · Easy · SPR
What is the solution to the equation $3x + 1 = 22$?

25. Linear equations in one variable · Easy · MC
What is the solution to the equation $4x + 2 = 46$?
A. 12
B. 10
C. 44
D. 11

26. Linear equations in one variable · Easy · MC
What is the solution to the equation $6x + 3 = 45$?
A. 7
B. 8
C. 6
D. 42

27. Linear equations in one variable · Easy · MC
What is the solution to the equation $3x + 13 = 37$?
A. 24
B. 7
C. 8
D. 9

28. Linear equations in one variable · Easy · SPR
What is the solution to the equation $5x + 10 = 65$?

29. Linear equations in one variable · Easy · MC
What is the solution to the equation $5x + 9 = 49$?
A. 40
B. 8
C. 7
D. 9

30. Linear equations in one variable · Easy · SPR
What is the solution to the equation $8x + 8 = 32$?

31. Linear equations in one variable · Easy · MC
What is the solution to the equation $5x + 14 = 49$?
A. 7
B. 6
C. 35
D. 8

32. Linear equations in one variable · Easy · MC
What is the solution to the equation $3x + 2 = 32$?
A. 10
B. 11
C. 9
D. 30

33. Linear equations in one variable · Easy · MC
What is the solution to the equation $2x + 13 = 33$?
A. 11
B. 9
C. 10
D. 20

34. Linear equations in one variable · Easy · MC
What is the solution to the equation $6x + 1 = 25$?
A. 24
B. 5
C. 3
D. 4

35. Linear equations in one variable · Easy · MC

What is the solution to the equation $5x + 9 = 64$?

A. 12

B. 11

C. 55

D. 10

36. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 3 = 36$?

A. 11

B. 12

C. 10

D. 33

37. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 2 = 17$?

A. 5

B. 15

C. 6

D. 4

38. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 15 = 39$?

A. 24

B. 9

C. 8

D. 7

39. Linear equations in one variable · Easy · MC

What is the solution to the equation $6x + 12 = 66$?

A. 8

B. 54

C. 10

D. 9

40. Linear equations in one variable · Easy · MC

What is the solution to the equation $2x + 7 = 29$?

A. 22

B. 10

C. 12

D. 11

41. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x + 6 = 6x + 26$?

A. 20

B. 16

C. -10

D. 10

42. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $7x - 4 = 2x + 16$?

43. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $6x - 1 = x + 54$?
A. 12
B. 11
C. -11
D. 55

44. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $4x - 5 = 3x - 8$?
A. -3
B. 0
C. 3
D. -2

45. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $4x + 8 = 2x + 26$?

46. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $5x - 1 = 2x + 20$?
A. 7
B. -7
C. 21
D. 9

47. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $5x - 2 = x + 30$?

48. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $4x = 3x + 4$?

49. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $7x + 10 = 5x + 20$?

50. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $4x + 1 = 2x + 21$?
A. 20
B. 12
C. -10
D. 10

51. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $8x + 7 = 5x + 28$?

52. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $8x - 3 = 4x - 15$?
A. 1
B. -3
C. 3
D. -12

53. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $7x = 5x - 8$?

54. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $5x + 10 = 4x + 17$?

55. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $7x + 10 = 2x - 15$?
A. 5
B. -25
C. -3
D. -5

56. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $3x - 1 = x - 7$?
A. -6
B. -3
C. -2
D. 3

57. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $3x - 8 = 2x - 13$?
A. -3
B. -5
C. 5
D. -4

58. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $8x - 5 = x + 2$?
A. 1
B. -1
C. 7
D. 2

59. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $5x - 6 = x - 22$?

60. Linear equations in one variable · Medium · SPR
What value of x satisfies the equation $4x - 3 = 2x - 9$?

61. Linear equations in one variable · Medium · MC
What value of x satisfies the equation $7x + 5 = 4x - 7$?
A. -4
B. 0
C. 4
D. -12

62. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $4x + 2 = 3x + 3$?

A. -1

B. 1

C. 4

D. 2

63. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $4x - 6 = 3x + 4$?

64. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $3x + 1 = 2x + 5$?

A. 5

B. 6

C. -4

D. 4

65. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x + 10 = 5x + 16$?

A. -2

B. 7

C. 2

D. 6

66. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $4x + 10 = 3x + 9$?

67. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x + 9 = 7x + 12$?

A. 3

B. -3

C. 10

D. 4

68. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $7x = 5x + 18$?

A. 18

B. 9

C. -9

D. 14

69. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $9x - 6 = 7x + 10$?

A. -8

B. 15

C. 16

D. 8

70. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $4x + 5 = x + 26$?

A. 7

B. 21

C. 8

D. -7

71. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $4x + 5 = 2x + 29$?

A. -12

B. 12

C. 24

D. 14

72. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x - 7 = 4x - 23$?

A. 4

B. -16

C. 0

D. -4

73. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $9x + 1 = 5x + 49$?

A. 12

B. 48

C. -12

D. 17

74. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x + 3 = 4x + 27$?

A. 6

B. 10

C. -6

D. 24

75. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $3x - 3 = x - 1$?

A. -2

B. 1

C. 2

D. -1

76. Linear equations in one variable · Hard · MC

What is the solution to $3(x + 4) = x + 30$?

A. -9

B. 9

C. 13

D. 12

77. Linear equations in one variable · Hard · MC
What is the solution to $3(x+3) = 6x - 9$?
A. 4
B. 6
C. -6
D. 9

78. Linear equations in one variable · Hard · MC
What is the solution to $3(x+5) = 6x + 6$?
A. $-\frac{1}{3}$
B. -3
C. 3
D. 6

79. Linear equations in one variable · Hard · MC
What is the solution to $4(x+4) = 3x + 13$?
A. 3
B. 9
C. 1
D. -3

80. Linear equations in one variable · Hard · MC
What is the solution to $4(x+1) = 2x$?
A. -2
B. -7
C. $-\frac{1}{2}$
D. 2

81. Linear equations in one variable · Hard · MC
What is the solution to $4(x+2) = 3x + 5$?
A. -10
B. 1
C. -3
D. 3

82. Linear equations in one variable · Hard · MC
What is the solution to $5(x+2) = x + 22$?
A. -3
B. 8
C. 3
D. 5

83. Linear equations in one variable · Hard · SPR
What is the solution to $3(x+6) = 6x + 27$?

84. Linear equations in one variable · Hard · MC
What is the solution to $2(x + 4) = 6x - 24$?
A. 8
B. -8
C. 10
D. 7

85. Linear equations in one variable · Hard · SPR
What is the solution to $5(x + 6) = 6x + 27$?

86. Linear equations in one variable · Hard · MC
What is the solution to $2(x + 2) = 3x + 1$?
A. -3
B. 3
C. 1
D. 5

87. Linear equations in one variable · Hard · MC
What is the solution to $4(x + 1) = 5x + 1$?
A. 0
B. 7
C. 3
D. -3

88. Linear equations in one variable · Hard · MC
What is the solution to $4(x + 3) = 5x + 9$?
A. 7
B. -6
C. -3
D. 3

89. Linear equations in one variable · Hard · MC
What is the solution to $2(x + 6) = 6x - 20$?
A. 10
B. -8
C. $\frac{13}{2}$
D. 8

90. Linear equations in one variable · Hard · MC
What is the solution to $5(x + 6) = 3x + 32$?
A. -1
B. 1
C. 6
D. 13

91. Linear equations in one variable · Hard · MC
What is the solution to $5(x+2) = 4x + 16$?
A. -6
B. 11
C. 6
D. 14

92. Linear equations in one variable · Hard · MC
What is the solution to $4(x+3) = 5x + 4$?
A. 12
B. 8
C. -8
D. -1

93. Linear equations in one variable · Hard · SPR
What is the solution to $5(x+1) = x + 37$?

94. Linear equations in one variable · Hard · MC
What is the solution to $4(x+3) = 2x + 22$?
A. $\frac{19}{2}$
B. 9
C. 5
D. -5

95. Linear equations in one variable · Hard · SPR
What is the solution to $3(x+1) = 2x + 11$?

96. Linear equations in one variable · Hard · MC
What is the solution to $4(x+1) = 2x + 8$?
A. $\frac{7}{2}$
B. 6
C. 2
D. -2

97. Linear equations in one variable · Hard · MC
What is the solution to $3(x+4) = x + 20$?
A. 4
B. 8
C. -4
D. 7

98. Linear equations in one variable · Hard · MC
What is the solution to $4(x+3) = 6x + 4$?
A. $-\frac{1}{2}$
B. 4
C. -4
D. 8

99. Linear equations in one variable · Hard · MC

What is the solution to $4(x+1) = 6x - 14$?

A. $\frac{15}{2}$

B. -9

C. 13

D. 9

100. Linear equations in one variable · Hard · MC

What is the solution to $2(x+1) = 4x - 14$?

A. 8

B. 10

C. $\frac{15}{2}$

D. -8

101. Number of solutions · Medium · MC

How many solutions does the equation $6x + 1 = 6x + 5$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

102. Number of solutions · Medium · MC

How many solutions does the equation $4(2x+1) = 8x + 4$ have?

A. Exactly two solutions

B. No solution

C. Infinitely many solutions

D. Exactly one solution

103. Number of solutions · Medium · MC

How many solutions does the equation $5(6x+6) = 30x + 30$ have?

A. Infinitely many solutions

B. Exactly one solution

C. Exactly two solutions

D. No solution

104. Number of solutions · Medium · MC

How many solutions does the equation $2(3x+7) = 6x + 14$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

105. Number of solutions · Medium · MC

How many solutions does the equation $2(5x+2) = 10x + 4$ have?

A. Exactly two solutions

B. No solution

C. Infinitely many solutions

D. Exactly one solution

106. Number of solutions · Medium · MC

How many solutions does the equation $4(5x + 5) = 20x + 20$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

107. Number of solutions · Medium · MC

How many solutions does the equation $7x + 7 = 7x + 10$ have?

A. No solution

B. Infinitely many solutions

C. Exactly one solution

D. Exactly two solutions

108. Number of solutions · Medium · MC

How many solutions does the equation $3(5x + 2) = 15x + 6$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

109. Number of solutions · Medium · MC

How many solutions does the equation $4x + 7 = 4x + 13$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

110. Number of solutions · Medium · MC

How many solutions does the equation $3x + 3 = 3x + 9$ have?

A. No solution

B. Infinitely many solutions

C. Exactly one solution

D. Exactly two solutions

111. Number of solutions · Medium · MC

How many solutions does the equation $4(8x + 2) = 32x + 8$ have?

A. Infinitely many solutions

B. Exactly one solution

C. Exactly two solutions

D. No solution

112. Number of solutions · Medium · MC

How many solutions does the equation $4x + 8 = 4x + 9$ have?

A. Exactly two solutions

B. No solution

C. Infinitely many solutions

D. Exactly one solution

113. Number of solutions · Hard · MC

How many solutions does the equation $2(5x + 7) = 10x + 14$ have?

A. Exactly one solution

B. Exactly two solutions

C. No solution

D. Infinitely many solutions

114. Number of solutions · Hard · MC

How many solutions does the equation $5x + 1 = 5x + 5$ have?

A. No solution

B. Infinitely many solutions

C. Exactly one solution

D. Exactly two solutions

115. Number of solutions · Hard · MC

How many solutions does the equation $4(3x + 4) = 12x + 16$ have?

A. Infinitely many solutions

B. Exactly one solution

C. Exactly two solutions

D. No solution

116. Number of solutions · Hard · MC

How many solutions does the equation $6x + 7 = 6x + 12$ have?

A. Exactly two solutions

B. No solution

C. Infinitely many solutions

D. Exactly one solution

117. Number of solutions · Hard · MC

How many solutions does the equation $5(3x + 1) = 15x + 5$ have?

A. No solution

B. Infinitely many solutions

C. Exactly one solution

D. Exactly two solutions

118. Number of solutions · Hard · MC

How many solutions does the equation $4x + 4 = 4x + 7$ have?

A. Infinitely many solutions

B. Exactly one solution

C. Exactly two solutions

D. No solution

119. Number of solutions · Hard · MC

How many solutions does the equation $4x + 6 = 4x + 12$ have?

A. No solution

B. Infinitely many solutions

C. Exactly one solution

D. Exactly two solutions

120. Number of solutions · Hard · MC

How many solutions does the equation $2(7x + 8) = 14x + 16$ have?

A. Infinitely many solutions

B. Exactly one solution

C. Exactly two solutions

D. No solution

121. Linear inequalities · Easy · MC

What is the solution set of $5x - 1 < 24$?

A. $x < 5$

B. $x > 5$

C. $x < 4$

D. $x \leq 5$

122. Linear inequalities · Easy · MC

What is the solution set of $5x - 8 < 42$?

A. $x < 9$

B. $x \leq 10$

C. $x < 10$

D. $x > 10$

123. Linear inequalities · Easy · MC

What is the solution set of $4x - 8 < 0$?

A. $x > 2$

B. $x < 1$

C. $x \leq 2$

D. $x < 2$

124. Linear inequalities · Easy · MC

What is the solution set of $4x - 2 < 26$?

A. $x < 6$

B. $x \leq 7$

C. $x < 7$

D. $x > 7$

125. Linear inequalities · Easy · MC

What is the solution set of $4x - 7 < 21$?

A. $x \leq 7$

B. $x < 7$

C. $x > 7$

D. $x < 6$

126. Linear inequalities · Easy · MC

What is the solution set of $5x - 7 < 18$?

A. $x < 4$

B. $x \leq 5$

C. $x < 5$

D. $x > 5$

127. Linear inequalities · Easy · MC
What is the solution set of $4x - 6 < 10$?
A. $x < 4$
B. $x > 4$
C. $x < 3$
D. $x \leq 4$

128. Linear inequalities · Easy · MC
What is the solution set of $5x - 5 < 10$?
A. $x < 2$
B. $x \leq 3$
C. $x < 3$
D. $x > 3$

129. Linear inequalities · Easy · MC
What is the solution set of $5x - 2 < 23$?
A. $x < 5$
B. $x > 5$
C. $x < 4$
D. $x \leq 5$

130. Linear inequalities · Easy · MC
What is the solution set of $4x - 5 < 27$?
A. $x > 8$
B. $x < 7$
C. $x \leq 8$
D. $x < 8$

131. Linear inequalities · Easy · MC
What is the solution set of $4x - 3 < 5$?
A. $x > 2$
B. $x < 1$
C. $x \leq 2$
D. $x < 2$

132. Linear inequalities · Easy · MC
What is the solution set of $4x - 2 < 38$?
A. $x \leq 10$
B. $x < 10$
C. $x > 10$
D. $x < 9$

133. Linear inequalities · Medium · MC
What is the solution set of $3x - 3 < 27$?
A. $x < 10$
B. $x > 10$
C. $x < 9$
D. $x \leq 10$

134. Linear inequalities · Medium · MC
What is the solution set of $3x - 4 < 14$?
A. $x \leq 6$
B. $x < 6$
C. $x > 6$
D. $x < 5$

135. Linear inequalities · Medium · MC
What is the solution set of $3x - 1 < 23$?
A. $x < 8$
B. $x > 8$
C. $x < 7$
D. $x \leq 8$

136. Linear inequalities · Medium · MC
What is the solution set of $3x - 1 < 20$?
A. $x < 7$
B. $x > 7$
C. $x < 6$
D. $x \leq 7$

137. Linear inequalities · Medium · MC
What is the solution set of $5x - 3 < 22$?
A. $x < 5$
B. $x > 5$
C. $x < 4$
D. $x \leq 5$

138. Linear inequalities · Medium · MC
What is the solution set of $5x - 6 < 24$?
A. $x > 6$
B. $x < 5$
C. $x \leq 6$
D. $x < 6$

139. Linear inequalities · Medium · MC
What is the solution set of $3x - 1 < 11$?
A. $x > 4$
B. $x < 3$
C. $x \leq 4$
D. $x < 4$

140. Linear inequalities · Medium · MC
What is the solution set of $2x - 8 < 0$?
A. $x \leq 4$
B. $x < 4$
C. $x > 4$
D. $x < 3$

141. Linear inequalities · Medium · MC
What is the solution set of $2x - 5 < 11$?
A. $x \leq 8$
B. $x < 8$
C. $x > 8$
D. $x < 7$

142. Linear inequalities · Medium · MC
What is the solution set of $4x - 3 < 33$?
A. $x < 9$
B. $x > 9$
C. $x < 8$
D. $x \leq 9$

143. Linear inequalities · Hard · MC
What is the solution set of $-5x + 5 < -20$?
A. $x > 5$
B. $x < 5$
C. $x > 6$
D. $x \geq 5$

144. Linear inequalities · Hard · MC
What is the solution set of $-5x + 1 < -9$?
A. $x \geq 2$
B. $x > 2$
C. $x < 2$
D. $x > 3$

145. Linear inequalities · Hard · MC
What is the solution set of $-2x + 5 < -9$?
A. $x > 7$
B. $x < 7$
C. $x > 8$
D. $x \geq 7$

146. Linear inequalities · Hard · MC
What is the solution set of $-5x + 3 < -32$?
A. $x \geq 7$
B. $x > 7$
C. $x < 7$
D. $x > 8$

147. Linear inequalities · Hard · MC
What is the solution set of $-2x + 2 < -18$?
A. $x < 10$
B. $x > 11$
C. $x \geq 10$
D. $x > 10$

148. Linear inequalities · Hard · MC

What is the solution set of $-3x + 2 < -10$?

A. $x \geq 4$

B. $x > 4$

C. $x < 4$

D. $x > 5$

149. Linear inequalities · Hard · MC

What is the solution set of $-4x + 3 < -17$?

A. $x \geq 5$

B. $x > 5$

C. $x < 5$

D. $x > 6$

150. Linear inequalities · Hard · MC

What is the solution set of $-4x + 4 < -20$?

A. $x > 7$

B. $x \geq 6$

C. $x > 6$

D. $x < 6$

151. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 1y > 5$?

A. $(2, 3)$

B. $(-1, 3)$

C. $(1, 1)$

D. $(0, 0)$

152. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 3y > 26$?

A. $(5, 6)$

B. $(2, 6)$

C. $(4, 4)$

D. $(0, 0)$

153. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 3y > 14$?

A. $(0, 0)$

B. $(5, 2)$

C. $(2, 2)$

D. $(4, 0)$

154. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 1y > 8$?

A. $(0, 0)$

B. $(4, 2)$

C. $(1, 2)$

D. $(3, 0)$

155. Linear inequalities · Medium · MC
Which point (x, y) is a solution of the inequality $3x + 1y > 12$?
A. $(3, 6)$
B. $(0, 6)$
C. $(2, 4)$
D. $(0, 0)$

156. Linear inequalities · Medium · MC
Which point (x, y) is a solution of the inequality $1x + 2y > 3$?
A. $(-1, 1)$
B. $(1, -1)$
C. $(0, 0)$
D. $(2, 1)$

157. Linear inequalities · Medium · MC
Which point (x, y) is a solution of the inequality $1x + 2y > 6$?
A. $(4, -1)$
B. $(0, 0)$
C. $(5, 1)$
D. $(2, 1)$

158. Linear inequalities · Medium · MC
Which point (x, y) is a solution of the inequality $2x + 1y > 13$?
A. $(4, 1)$
B. $(6, -1)$
C. $(0, 0)$
D. $(7, 1)$

159. Linear inequalities · Medium · MC
Which point (x, y) is a solution of the inequality $2x + 1y > 6$?
A. $(2, 4)$
B. $(-1, 4)$
C. $(1, 2)$
D. $(0, 0)$

160. Linear inequalities · Medium · MC
Which point (x, y) is a solution of the inequality $2x + 1y > 7$?
A. $(2, 1)$
B. $(0, 0)$
C. $(3, 3)$
D. $(0, 3)$

161. Linear equations in two variables · Easy · MC
What is the slope of the line through $(1, 0)$ and $(5, 6)$?
A. $\frac{2}{3}$
B. 6
C. $\frac{3}{2}$
D. $-\frac{3}{2}$

162. Linear equations in two variables · Easy · MC
What is the slope of the line through $(2, 2)$ and $(5, 6)$?
A. $-\frac{4}{3}$
B. $\frac{3}{4}$
C. $\frac{4}{3}$
D. 4

163. Linear equations in two variables · Easy · MC
What is the slope of the line through $(1, 4)$ and $(3, 7)$?
A. $-\frac{3}{2}$
B. 3
C. $\frac{2}{3}$
D. $\frac{3}{2}$

164. Linear equations in two variables · Easy · MC
What is the slope of the line through $(3, 3)$ and $(4, 7)$?
A. $\frac{1}{4}$
B. 4
C. 1
D. -4

165. Linear equations in two variables · Easy · SPR
What is the slope of the line through $(0, 2)$ and $(2, 8)$?

166. Linear equations in two variables · Easy · MC
What is the slope of the line through $(0, 0)$ and $(1, 6)$?
A. 1
B. $\frac{1}{6}$
C. -6
D. 6

167. Linear equations in two variables · Easy · SPR
What is the slope of the line through $(2, 4)$ and $(4, 5)$?

168. Linear equations in two variables · Easy · MC
What is the slope of the line through $(3, 2)$ and $(4, 3)$?
A. 1
B. 0
C. 2
D. -1

169. Linear equations in two variables · Easy · MC
What is the slope of the line through $(3, 2)$ and $(5, 6)$?
A. 2
B. $\frac{1}{2}$
C. 4
D. -2

170. Linear equations in two variables · Easy · MC
What is the slope of the line through $(3, 0)$ and $(5, 4)$?
A. $\frac{1}{2}$
B. -2
C. 4
D. 2

171. Linear equations in two variables · Easy · MC
What is the slope of the line through $(1, 4)$ and $(4, 7)$?
A. 3
B. -1
C. 1
D. 2

172. Linear equations in two variables · Easy · SPR
What is the slope of the line through $(4, 3)$ and $(5, 6)$?

173. Linear equations in two variables · Easy · MC
What is the slope of the line through $(1, 3)$ and $(4, 6)$?
A. -1
B. 2
C. 3
D. 1

174. Linear equations in two variables · Easy · MC
What is the slope of the line through $(0, 0)$ and $(1, 3)$?
A. 1
B. $\frac{1}{3}$
C. 3
D. -3

175. Linear equations in two variables · Easy · SPR
What is the slope of the line through $(3, 0)$ and $(6, 2)$?

176. Linear equations in two variables · Easy · MC
What is the slope of the line through $(0, 0)$ and $(4, 2)$?
A. 4
B. $\frac{1}{2}$
C. $-\frac{1}{2}$
D. 2

177. Linear equations in two variables · Easy · MC
What is the slope of the line through $(3, 0)$ and $(7, 4)$?
A. -1
B. 4
C. 1
D. 2

178. Linear equations in two variables · Easy · SPR
What is the slope of the line through $(1, 1)$ and $(2, 4)$?

179. Linear equations in two variables · Easy · SPR
What is the slope of the line through $(4, 1)$ and $(6, 3)$?

180. Linear equations in two variables · Easy · MC
What is the slope of the line through $(0, 3)$ and $(2, 4)$?
A. 1
B. 2
C. $\frac{1}{2}$
D. $-\frac{1}{2}$

181. Linear equations in two variables · Easy · MC
What is the slope of the line through $(4, 0)$ and $(6, 1)$?
A. $-\frac{1}{2}$
B. $\frac{1}{2}$
C. 2
D. 1

182. Linear equations in two variables · Easy · SPR
What is the slope of the line through $(1, 0)$ and $(3, 6)$?

183. Linear functions · Medium · MC
The line with equation $y = 2x + 3$ passes through the point $(3, k)$. What is the value of k?
A. 6
B. 9
C. 3
D. 5

184. Linear functions · Medium · MC
The line with equation $y = 4x - 1$ passes through the point $(1, k)$. What is the value of k?
A. 3
B. 0
C. 5
D. -3

185. Linear functions · Medium · MC
The line with equation $y = -\frac{1}{2}x + 6$ passes through the point $(2, k)$. What is the value of k?
A. -7
B. 8
C. 5
D. $\frac{11}{2}$

186. Linear functions · Medium · MC
The line with equation $y = 2x - 4$ passes through the point $(-1, k)$. What is the value of k?
A. 2
B. -6
C. -5
D. -2

187. Linear functions · Medium · MC

The line with equation $y = 2x + 2$ passes through the point $(2, k)$. What is the value of k?

A. 6

B. -6

C. 2

D. 4

188. Linear functions · Medium · SPR

The line with equation $y = 4x$ passes through the point $(4, k)$. What is the value of k?

189. Linear functions · Medium · MC

The line with equation $y = 3x + 5$ passes through the point $(0, k)$. What is the value of k?

A. -5

B. 8

C. 5

D. 6

190. Linear functions · Medium · MC

The line with equation $y = -x + 4$ passes through the point $(5, k)$. What is the value of k?

A. -1

B. -9

C. 3

D. 9

191. Linear functions · Medium · MC

The line with equation $y = 2x$ passes through the point $(4, k)$. What is the value of k?

A. -8

B. 2

C. 8

D. 4

192. Linear functions · Medium · MC

The line with equation $y = 2x - 4$ passes through the point $(4, k)$. What is the value of k?

A. -2

B. 12

C. 0

D. 4

193. Linear functions · Medium · MC

The line with equation $y = x - 1$ passes through the point $(-2, k)$. What is the value of k?

A. 0

B. -1

C. 3

D. -3

194. Linear functions · Medium · MC

The line with equation $y = -3x + 4$ passes through the point $(3, k)$. What is the value of k?

A. 1

B. -13

C. -5

D. 7

195. Linear functions · Medium · MC

The line with equation $y = 4x + 4$ passes through the point $(0, k)$. What is the value of k?

A. -4

B. 5

C. 8

D. 4

196. Linear functions · Medium · SPR

The line with equation $y = 3x + 3$ passes through the point $(2, k)$. What is the value of k?

197. Linear functions · Medium · MC

The line with equation $y = -2x + 1$ passes through the point $(4, k)$. What is the value of k?

A. -9

B. 5

C. -7

D. -1

198. Linear functions · Medium · SPR

The line with equation $y = x + 3$ passes through the point $(5, k)$. What is the value of k?

199. Linear functions · Medium · MC

The line with equation $y = \frac{3}{2}x - 1$ passes through the point $(-2, k)$. What is the value of k?

A. -2

B. $\frac{1}{2}$

C. -3

D. -4

200. Linear functions · Medium · SPR

The line with equation $y = 2x + 6$ passes through the point $(-1, k)$. What is the value of k?

201. Linear functions · Medium · MC

The line with equation $y = 2x + 3$ passes through the point $(5, k)$. What is the value of k?

A. 5

B. 13

C. 7

D. 8

202. Linear functions · Medium · MC

The line with equation $y = x + 2$ passes through the point $(-2, k)$. What is the value of k?

A. 1

B. 3

C. -4

D. 0

203. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{3}{2}x + 3$. What is the slope of the perpendicular line?

A. $\frac{3}{2}$

B. $-\frac{2}{3}$

C. $-\frac{3}{2}$

D. $\frac{2}{3}$

204. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{2}{3}x - 3$. What is the slope of the perpendicular line?

A. $-\frac{3}{2}$

B. $\frac{2}{3}$

C. $-\frac{2}{3}$

D. $\frac{3}{2}$

205. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = x - 2$. What is the slope of the perpendicular line?

A. 2

B. 0

C. 1

D. -1

206. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 4x + 3$. What is the slope of the perpendicular line?

A. -4

B. 4

C. $-\frac{1}{4}$

D. $\frac{1}{4}$

207. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{3}{2}x - 1$. What is the slope of the perpendicular line?

A. $\frac{3}{2}$

B. $-\frac{3}{2}$

C. $\frac{2}{3}$

D. $-\frac{2}{3}$

208. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{1}{2}x + 1$. What is the slope of the perpendicular line?

A. $\frac{1}{2}$

B. 2

C. $-\frac{1}{2}$

D. -2

209. Linear equations in two variables · Hard · SPR

A line is perpendicular to the line $y = 3x + 2$. What is the slope of the perpendicular line?

210. Linear equations in two variables · Hard · SPR

A line is perpendicular to the line $y = 2x - 1$. What is the slope of the perpendicular line?

211. Linear equations in two variables · Hard · SPR

A line is perpendicular to the line $y = \frac{1}{3}x + 4$. What is the slope of the perpendicular line?

212. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = x$. What is the slope of the perpendicular line?

A. 2

B. -1

C. 0

D. 1

213. Linear equations in two variables · Hard · SPR

A line is perpendicular to the line $y = 3x$. What is the slope of the perpendicular line?

214. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{3}{2}x + 4$. What is the slope of the perpendicular line?

A. $-\frac{2}{3}$

B. $\frac{3}{2}$

C. $\frac{2}{3}$

D. $-\frac{3}{2}$

215. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 3x + 1$. What is the slope of the perpendicular line?

A. -3

B. $\frac{1}{3}$

C. 3

D. $-\frac{1}{3}$

216. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 2x - 3$. What is the slope of the perpendicular line?

A. -2

B. $\frac{1}{2}$

C. $-\frac{1}{2}$

D. 2

217. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{4}{3}x - 3$. What is the slope of the perpendicular line?

A. $\frac{3}{4}$

B. $-\frac{4}{3}$

C. $\frac{4}{3}$

D. $-\frac{3}{4}$

218. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 4x - 2$. What is the slope of the perpendicular line?

A. $-\frac{1}{4}$

B. -4

C. $\frac{1}{4}$

D. 4

219. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{1}{2}x + 3$. What is the slope of the perpendicular line?

A. 2

B. $-\frac{1}{2}$

C. -2

D. $\frac{1}{2}$

220. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 2x + 3$. What is the slope of the perpendicular line?

A. $\frac{1}{2}$

B. $-\frac{1}{2}$

C. -2

D. 2

221. Linear functions · Easy · MC

A phone battery is at 200% and loses 12 percentage points each hour during a call. Which equation gives p, the amount remaining after h hours, assuming the rate stays constant?

A. $p = 12h - 200$

B. $p = 200 + 12h$

C. $p = 12 - 200h$

D. $p = 200 - 12h$

222. Linear functions · Easy · MC

A student has \$150 saved and spends \$3 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 150 + 3w$

B. $d = 3 - 150w$

C. $d = 150 - 3w$

D. $d = 3w - 150$

223. Linear functions · Easy · MC

A tank contains 80 gallons of water and drains at a constant rate of 4 gallons per minute. Which equation gives g, the amount remaining after m minutes, assuming the rate stays constant?

A. $g = 4m - 80$

B. $g = 80 + 4m$

C. $g = 4 - 80m$

D. $g = 80 - 4m$

224. Linear functions · Easy · MC

A phone battery is at 60% and loses 3 percentage points each hour during a call. Which equation gives p, the amount remaining after h hours, assuming the rate stays constant?

A. $p = 3 - 60h$

B. $p = 60 - 3h$

C. $p = 3h - 60$

D. $p = 60 + 3h$

225. Linear functions · Easy · MC

A phone battery is at 80% and loses 3 percentage points each hour during a call. Which equation gives p, the amount remaining after h hours, assuming the rate stays constant?

A. $p = 3h - 80$

B. $p = 80 + 3h$

C. $p = 3 - 80h$

D. $p = 80 - 3h$

226. Linear functions · Easy · MC

A student has \$60 saved and spends \$3 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 60 + 3w$

B. $d = 3 - 60w$

C. $d = 60 - 3w$

D. $d = 3w - 60$

227. Linear functions · Easy · MC

A phone battery is at 80% and loses 6 percentage points each hour during a call. Which equation gives p, the amount remaining after h hours, assuming the rate stays constant?

A. $p = 6h - 80$

B. $p = 80 + 6h$

C. $p = 6 - 80h$

D. $p = 80 - 6h$

228. Linear functions · Easy · MC

A tank contains 150 gallons of water and drains at a constant rate of 8 gallons per minute. Which equation gives g, the amount remaining after m minutes, assuming the rate stays constant?

A. $g = 150 + 8m$

B. $g = 8 - 150m$

C. $g = 150 - 8m$

D. $g = 8m - 150$

229. Linear functions · Easy · MC

A student has \$60 saved and spends \$4 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 60 - 4w$

B. $d = 4w - 60$

C. $d = 60 + 4w$

D. $d = 4 - 60w$

230. Linear functions · Easy · MC

A student has \$120 saved and spends \$8 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 8 - 120w$

B. $d = 120 - 8w$

C. $d = 8w - 120$

D. $d = 120 + 8w$

231. Linear functions · Easy · MC

A student has \$60 saved and spends \$6 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 6 - 60w$

B. $d = 60 - 6w$

C. $d = 6w - 60$

D. $d = 60 + 6w$

232. Linear functions · Easy · MC

A student has \$150 saved and spends \$12 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 12 - 150w$

B. $d = 150 - 12w$

C. $d = 12w - 150$

D. $d = 150 + 12w$

233. Linear functions · Medium · MC

A student has \$60 saved and spends \$10 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 10 - 60w$

B. $d = 60 - 10w$

C. $d = 10w - 60$

D. $d = 60 + 10w$

234. Linear functions · Medium · MC

A student has \$120 saved and spends \$12 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 120 - 12w$

B. $d = 12w - 120$

C. $d = 120 + 12w$

D. $d = 12 - 120w$

235. Linear functions · Medium · MC

A phone battery is at 120% and loses 6 percentage points each hour during a call. Which equation gives p, the amount remaining after h hours, assuming the rate stays constant?

A. $p = 6 - 120h$

B. $p = 120 - 6h$

C. $p = 6h - 120$

D. $p = 120 + 6h$

236. Linear functions · Medium · MC

A phone battery is at 200% and loses 5 percentage points each hour during a call. Which equation gives p, the amount remaining after h hours, assuming the rate stays constant?

A. $p = 5 - 200h$

B. $p = 200 - 5h$

C. $p = 5h - 200$

D. $p = 200 + 5h$

237. Linear functions · Medium · MC

A student has \$40 saved and spends \$5 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 40 - 5w$

B. $d = 5w - 40$

C. $d = 40 + 5w$

D. $d = 5 - 40w$

238. Linear functions · Medium · MC

A tank contains 40 gallons of water and drains at a constant rate of 6 gallons per minute. Which equation gives g, the amount remaining after m minutes, assuming the rate stays constant?

A. $g = 40 - 6m$

B. $g = 6m - 40$

C. $g = 40 + 6m$

D. $g = 6 - 40m$

239. Linear functions · Medium · MC

A tank contains 200 gallons of water and drains at a constant rate of 4 gallons per minute. Which equation gives g, the amount remaining after m minutes, assuming the rate stays constant?

A. $g = 4m - 200$

B. $g = 200 + 4m$

C. $g = 4 - 200m$

D. $g = 200 - 4m$

240. Linear functions · Medium · MC

A phone battery is at 120% and loses 8 percentage points each hour during a call. Which equation gives p, the amount remaining after h hours, assuming the rate stays constant?

A. $p = 120 + 8h$

B. $p = 8 - 120h$

C. $p = 120 - 8h$

D. $p = 8h - 120$

241. Linear functions · Medium · MC

A tank contains 80 gallons of water and drains at a constant rate of 3 gallons per minute. Which equation gives g, the amount remaining after m minutes, assuming the rate stays constant?

A. $g = 3 - 80m$

B. $g = 80 - 3m$

C. $g = 3m - 80$

D. $g = 80 + 3m$

242. Linear functions · Medium · MC

A tank contains 60 gallons of water and drains at a constant rate of 3 gallons per minute. Which equation gives g, the amount remaining after m minutes, assuming the rate stays constant?

A. $g = 60 - 3m$

B. $g = 3m - 60$

C. $g = 60 + 3m$

D. $g = 3 - 60m$

243. Linear functions · Medium · MC

A tank contains 240 gallons of water and drains at a constant rate of 8 gallons per minute. Which equation gives g, the amount remaining after m minutes, assuming the rate stays constant?

A. $g = 8m - 240$

B. $g = 240 + 8m$

C. $g = 8 - 240m$

D. $g = 240 - 8m$

244. Linear functions · Medium · MC

A student has \$60 saved and spends \$8 each week. Which equation gives d, the amount remaining after w weeks, assuming the rate stays constant?

A. $d = 60 - 8w$

B. $d = 8w - 60$

C. $d = 60 + 8w$

D. $d = 8 - 60w$

245. Linear functions · Medium · MC

A phone battery is at 120% and loses 4 percentage points each hour during a call. Which equation gives p, the amount remaining after h hours, assuming the rate stays constant?

A. $p = 120 - 4h$

B. $p = 4h - 120$

C. $p = 120 + 4h$

D. $p = 4 - 120h$

246. Systems of linear equations · Easy · SPR

The solution of the system $x + y = 1$ and $x - y = 3$ is (x, y). What is the value of x?

247. Systems of linear equations · Easy · SPR

The solution of the system $x + y = -7$ and $x - y = 1$ is (x, y). What is the value of x?

248. Systems of linear equations · Easy · MC

The solution of the system $x + y = -5$ and $x - y = -3$ is (x, y). What is the value of x?

A. -3

B. -4

C. -5

D. -1

249. Systems of linear equations · Easy · MC

The solution of the system $x + y = 7$ and $x - y = 3$ is (x, y). What is the value of x?

A. 5

B. 3

C. 2

D. 7

250. Systems of linear equations · Easy · MC

The solution of the system $x + y = 5$ and $x - y = -3$ is (x, y). What is the value of x?

A. 5

B. 1

C. -3

D. 4

251. Systems of linear equations · Easy · MC

The solution of the system $x + y = -6$ and $x - y = 0$ is (x, y). What is the value of x?

A. -6

B. -2

C. -3

D. 0

252. Systems of linear equations · Easy · MC

The solution of the system $x + y = 15$ and $x - y = 1$ is (x, y). What is the value of x?

A. 7

B. 1

C. 15

D. 8

253. Systems of linear equations · Easy · MC

The solution of the system $x + y = 0$ and $x - y = 0$ is (x, y). What is the value of x?

A. 0

B. 1

C. -1

D. 2

254. Systems of linear equations · Easy · MC

The solution of the system $x + y = 3$ and $x - y = -1$ is (x, y). What is the value of x?

A. -1

B. 2

C. 1

D. 3

255. Systems of linear equations · Easy · MC

The solution of the system $x + y = -1$ and $x - y = 3$ is (x, y). What is the value of x?

A. 1

B. 3

C. -1

D. -2

256. Systems of linear equations · Easy · MC

The solution of the system $x + y = 0$ and $x - y = -2$ is (x, y). What is the value of x?

A. 1

B. -2

C. -1

D. 0

257. Systems of linear equations · Easy · MC

The solution of the system $x + y = -5$ and $x - y = 3$ is (x, y). What is the value of x?

A. -5

B. -1

C. 3

D. -4

258. Systems of linear equations · Easy · MC

The solution of the system $x + y = 4$ and $x - y = 2$ is (x, y). What is the value of x?

A. 1

B. 4

C. 3

D. 2

259. Systems of linear equations · Easy · MC

The solution of the system $x + y = -2$ and $x - y = -6$ is (x, y). What is the value of x?

A. -2

B. 2

C. -6

D. -4

260. Systems of linear equations · Easy · MC

The solution of the system $x + y = -4$ and $x - y = 0$ is (x, y). What is the value of x?

A. -4

B. -1

C. 0

D. -2

261. Systems of linear equations · Easy · MC

The solution of the system $x + y = 12$ and $x - y = 0$ is (x, y). What is the value of x?

A. 7

B. 12

C. 6

D. 0

262. Systems of linear equations · Easy · MC

The solution of the system $x + y = 5$ and $x - y = 1$ is (x, y). What is the value of x?

A. 2

B. 1

C. 3

D. 5

263. Systems of linear equations · Easy · MC

The solution of the system $x + y = 11$ and $x - y = 1$ is (x, y). What is the value of x?

A. 1

B. 6

C. 11

D. 5

264. Systems of linear equations · Easy · SPR

The solution of the system $x + y = 2$ and $x - y = -4$ is (x, y). What is the value of x?

265. Systems of linear equations · Easy · MC

The solution of the system $x + y = 11$ and $x - y = 5$ is (x, y). What is the value of x?

A. 11

B. 8

C. 3

D. 5

266. Systems of linear equations · Medium · MC

What is the value of x in the solution of $2x + 5y = 36$ and $x + 2y = 15$?

A. -3

B. 36

C. 6

D. 3

267. Systems of linear equations · Medium · MC

What is the value of x in the solution of $5x + 2y = 6$ and $x - 2y = -18$?

A. -10

B. -2

C. 8

D. 6

268. Systems of linear equations · Medium · SPR

What is the value of x in the solution of $5x + 2y = 25$ and $x - y = 5$?

269. Systems of linear equations · Medium · MC

What is the value of x in the solution of $3x + 4y = 16$ and $x - y = -11$?

A. -4

B. 16

C. 7

D. -11

270. Systems of linear equations · Medium · MC

What is the value of x in the solution of $2x + 5y = 51$ and $x - y = 1$?

A. 7

B. 8

C. 1

D. 51

271. Systems of linear equations · Medium · MC

What is the value of x in the solution of $3x + 2y = -10$ and $x - 2y = -6$?

A. -5

B. 1

C. -10

D. -4

272. Systems of linear equations · Medium · MC
What is the value of x in the solution of $5x + 2y = 6$ and $x - 2y = 6$?
A. 2
B. -2
C. 6
D. 4

273. Systems of linear equations · Medium · MC
What is the value of x in the solution of $2x + 3y = 6$ and $2x - y = 6$?
A. 6
B. 4
C. 0
D. 3

274. Systems of linear equations · Medium · MC
What is the value of x in the solution of $4x + y = -4$ and $x + 2y = 6$?
A. 4
B. -2
C. -4
D. -6

275. Systems of linear equations · Medium · MC
What is the value of x in the solution of $2x + 3y = 30$ and $x - y = -5$?
A. 8
B. 30
C. -5
D. 3

276. Systems of linear equations · Medium · SPR
What is the value of x in the solution of $3x + 2y = 32$ and $2x - y = 12$?

277. Systems of linear equations · Medium · SPR
What is the value of x in the solution of $2x + 5y = 54$ and $x - y = -1$?

278. Systems of linear equations · Medium · SPR
What is the value of x in the solution of $3x + 4y = 9$ and $x - 2y = 13$?

279. Systems of linear equations · Medium · MC
What is the value of x in the solution of $2x + 5y = 40$ and $x + 2y = 16$?
A. 8
B. -8
C. 40
D. 0

280. Systems of linear equations · Medium · MC
What is the value of x in the solution of $2x + 5y = 18$ and $x + 2y = 8$?
A. 4
B. 18
C. 5
D. 2

281. Systems of linear equations · Medium · MC
What is the value of x in the solution of $2x + 3y = 9$ and $2x - y = 13$?
A. 7
B. -1
C. 6
D. 9

282. Systems of linear equations · Medium · MC
What is the value of x in the solution of $3x + 4y = 20$ and $x - y = 9$?
A. 9
B. 8
C. -1
D. 20

283. Systems of linear equations · Medium · MC
What is the value of x in the solution of $4x + y = -8$ and $3x + y = -4$?
A. -8
B. -12
C. -4
D. 8

284. Systems of linear equations · Medium · SPR
What is the value of x in the solution of $5x + 2y = 19$ and $x - 2y = -13$?

285. Systems of linear equations · Medium · SPR
What is the value of x in the solution of $2x + 5y = 37$ and $x + 2y = 15$?

286. Systems of linear equations · Hard · MC
What is the value of $x + y$ in the solution of $3x + 2y = 29$ and $x + 2y = 15$?
A. 4
B. 3
C. 7
D. 11

287. Systems of linear equations · Hard · SPR
What is the value of $x + y$ in the solution of $5x + 2y = 11$ and $2x - y = -10$?

288. Systems of linear equations · Hard · MC
What is the value of $x + y$ in the solution of $2x + 3y = 36$ and $x - 2y = -10$?
A. 14
B. -2
C. 6
D. 8

289. Systems of linear equations · Hard · MC
What is the value of $x + y$ in the solution of $5x + 2y = 9$ and $2x - y = 9$?
A. 6
B. 3
C. 0
D. -3

290. Systems of linear equations · Hard · MC
What is the value of $x + y$ in the solution of $2x + 5y = 22$ and $x - 2y = 2$?
A. 8
B. 6
C. 2
D. 4

291. Systems of linear equations · Hard · MC
What is the value of $x + y$ in the solution of $3x + 2y = 21$ and $x + 2y = 11$?
A. 8
B. 5
C. 2
D. 3

292. Systems of linear equations · Hard · MC
What is the value of $x + y$ in the solution of $5x + 2y = 37$ and $3x + y = 22$?
A. 7
B. 1
C. 8
D. 6

293. Systems of linear equations · Hard · SPR
What is the value of $x + y$ in the solution of $2x + 3y = 22$ and $x - y = 1$?

294. Systems of linear equations · Hard · MC
What is the value of $x + y$ in the solution of $2x + 3y = 24$ and $x - 2y = -2$?
A. 10
B. 2
C. 4
D. 6

295. Systems of linear equations · Hard · SPR
What is the value of $x + y$ in the solution of $4x + y = 10$ and $2x - y = -4$?

296. Systems of linear equations · Hard · MC
What is the value of $x + y$ in the solution of $4x + y = 20$ and $3x + y = 14$?
A. 2
B. 10
C. -4
D. 6

297. Systems of linear equations · Hard · MC
What is the value of $x + y$ in the solution of $2x + 3y = 17$ and $x - 2y = -2$?
A. 4
B. 7
C. 3
D. 1

298. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $2x + 5y = -20$ and $x - 2y = 8$?

A. -4

B. 4

C. 0

D. -20

299. Systems of linear equations · Hard · SPR

What is the value of $x + y$ in the solution of $4x + y = 27$ and $3x + y = 20$?

300. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $2x + 5y = 13$ and $2x - y = -5$?

A. 3

B. -1

C. 2

D. -4

301. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x - 2$ and $y = 4x + 2$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. None

302. Systems of linear equations · Medium · MC

How many solutions does the system $y = 5x + 3$ and $3y = 15x + 9$ have?

A. None

B. Exactly one

C. Exactly two

D. Infinitely many

303. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x$ and $y = 4x + 2$ have?

A. Exactly two

B. Infinitely many

C. None

D. Exactly one

304. Systems of linear equations · Medium · MC

How many solutions does the system $y = 2x + 4$ and $2y = 4x + 8$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. None

305. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x + 3$ and $3y = 12x + 9$ have?

A. Exactly two

B. Infinitely many

C. None

D. Exactly one

306. Systems of linear equations · Medium · MC

How many solutions does the system $y = 3x + 5$ and $y = 3x + 7$ have?

A. None

B. Exactly one

C. Exactly two

D. Infinitely many

307. Systems of linear equations · Medium · MC

How many solutions does the system $y = 5x + 5$ and $y = 5x + 9$ have?

A. Exactly two

B. Infinitely many

C. None

D. Exactly one

308. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x + 4$ and $y = 4x + 8$ have?

A. None

B. Exactly one

C. Exactly two

D. Infinitely many

309. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x + 2$ and $y = 4x + 6$ have?

A. None

B. Exactly one

C. Exactly two

D. Infinitely many

310. Systems of linear equations · Medium · MC

How many solutions does the system $y = 3x + 1$ and $3y = 9x + 3$ have?

A. Exactly two

B. Infinitely many

C. None

D. Exactly one

311. Solve for a variable · Easy · MC

The perimeter of a rectangle is 46 and the width is 10. What is the length?

A. 23

B. 10

C. 13

D. 36

312. Solve for a variable · Easy · SPR
The perimeter of a rectangle is 24 and the width is 7. What is the length?

313. Solve for a variable · Easy · SPR
The perimeter of a rectangle is 42 and the width is 6. What is the length?

314. Solve for a variable · Easy · SPR
The perimeter of a rectangle is 16 and the width is 3. What is the length?

315. Solve for a variable · Easy · MC
The perimeter of a rectangle is 44 and the width is 10. What is the length?
A. 22
B. 12
C. 34
D. 10

316. Solve for a variable · Easy · MC
The perimeter of a rectangle is 34 and the width is 10. What is the length?
A. 7
B. 10
C. 24
D. 17

317. Solve for a variable · Easy · SPR
The perimeter of a rectangle is 30 and the width is 5. What is the length?

318. Solve for a variable · Easy · MC
The perimeter of a rectangle is 22 and the width is 7. What is the length?
A. 11
B. 15
C. 4
D. 7

319. Solve for a variable · Easy · MC
The perimeter of a rectangle is 44 and the width is 7. What is the length?
A. 22
B. 37
C. 15
D. 7

320. Solve for a variable · Easy · MC
The perimeter of a rectangle is 34 and the width is 3. What is the length?
A. 14
B. 17
C. 3
D. 31

321. Solve for a variable · Easy · MC
The perimeter of a rectangle is 28 and the width is 4. What is the length?
A. 14
B. 4
C. 24
D. 10

322. Solve for a variable · Easy · MC
The perimeter of a rectangle is 42 and the width is 9. What is the length?
A. 21
B. 33
C. 9
D. 12

323. Solve for a variable · Easy · MC
The perimeter of a rectangle is 14 and the width is 2. What is the length?
A. 7
B. 12
C. 2
D. 5

324. Solve for a variable · Easy · MC
The perimeter of a rectangle is 28 and the width is 6. What is the length?
A. 6
B. 22
C. 14
D. 8

325. Solve for a variable · Easy · SPR
The perimeter of a rectangle is 18 and the width is 5. What is the length?

326. Solve for a variable · Medium · SPR
The perimeter of a rectangle is 22 and the width is 4. What is the length?

327. Solve for a variable · Medium · MC
The perimeter of a rectangle is 24 and the width is 6. What is the length?
A. 9
B. 12
C. 6
D. 18

328. Solve for a variable · Medium · MC
The perimeter of a rectangle is 32 and the width is 6. What is the length?
A. 16
B. 10
C. 26
D. 6

329. Solve for a variable · Medium · MC
The perimeter of a rectangle is 26 and the width is 6. What is the length?
A. 20
B. 7
C. 6
D. 13

330. Solve for a variable · Medium · MC
The perimeter of a rectangle is 26 and the width is 4. What is the length?
A. 13
B. 4
C. 22
D. 9

331. Solve for a variable · Medium · SPR
The perimeter of a rectangle is 26 and the width is 9. What is the length?

332. Solve for a variable · Medium · MC
The perimeter of a rectangle is 28 and the width is 2. What is the length?
A. 2
B. 26
C. 14
D. 12

333. Solve for a variable · Medium · SPR
The perimeter of a rectangle is 36 and the width is 4. What is the length?

334. Solve for a variable · Medium · SPR
The perimeter of a rectangle is 30 and the width is 4. What is the length?

335. Solve for a variable · Medium · MC
The perimeter of a rectangle is 26 and the width is 5. What is the length?
A. 5
B. 13
C. 8
D. 21

336. Solve for a variable · Medium · MC
The perimeter of a rectangle is 22 and the width is 6. What is the length?
A. 16
B. 5
C. 11
D. 6

337. Solve for a variable · Medium · MC
The perimeter of a rectangle is 38 and the width is 7. What is the length?
A. 31
B. 12
C. 19
D. 7

338. Solve for a variable · Medium · MC

The perimeter of a rectangle is 30 and the width is 2. What is the length?

A. 13

B. 2

C. 28

D. 15

339. Solve for a variable · Medium · MC

The perimeter of a rectangle is 22 and the width is 5. What is the length?

A. 6

B. 5

C. 11

D. 17

340. Solve for a variable · Medium · SPR

The perimeter of a rectangle is 44 and the width is 9. What is the length?

341. Solve for a variable · Hard · MC

A car travels 36 miles in 2 hours at a constant speed. What is the speed in miles per hour?

A. 72

B. 38

C. 2

D. 18

342. Solve for a variable · Hard · MC

A car travels 48 miles in 6 hours at a constant speed. What is the speed in miles per hour?

A. 288

B. 8

C. 6

D. 54

343. Solve for a variable · Hard · MC

A car travels 28 miles in 2 hours at a constant speed. What is the speed in miles per hour?

A. 14

B. 56

C. 2

D. 30

344. Solve for a variable · Hard · MC

A car travels 144 miles in 6 hours at a constant speed. What is the speed in miles per hour?

A. 150

B. 24

C. 864

D. 6

345. Solve for a variable · Hard · SPR

A car travels 33 miles in 3 hours at a constant speed. What is the speed in miles per hour?

346. Solve for a variable · Hard · MC

A car travels 50 miles in 5 hours at a constant speed. What is the speed in miles per hour?

A. 250

B. 55

C. 10

D. 5

347. Solve for a variable · Hard · SPR

A car travels 36 miles in 3 hours at a constant speed. What is the speed in miles per hour?

348. Solve for a variable · Hard · MC

A car travels 65 miles in 5 hours at a constant speed. What is the speed in miles per hour?

A. 70

B. 325

C. 13

D. 5

349. Solve for a variable · Hard · MC

A car travels 95 miles in 5 hours at a constant speed. What is the speed in miles per hour?

A. 475

B. 100

C. 19

D. 5

350. Solve for a variable · Hard · SPR

A car travels 50 miles in 2 hours at a constant speed. What is the speed in miles per hour?

Advanced Math

351. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x+6)(x+5)$?

A. $x^2+11x+30$

B. $x^2+30x+11$

C. $x^2+11x+12$

D. $x^2-11x+30$

352. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x+1)(x+7)$?

A. x^2-8x+7

B. x^2+8x+7

C. x^2+7x+8

D. x^2+8x+9

353. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x+8)(x+6)$?

A. $x^2+14x+48$

B. $x^2+48x+14$

C. $x^2+14x+15$

D. $x^2-14x+48$

354. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x+8)(x+5)$?

A. $x^2+13x+14$

B. $x^2-13x+40$

C. $x^2+13x+40$

D. $x^2+40x+13$

355. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x+6)(x+2)$?

A. $x^2+8x+12$

B. $x^2+12x+8$

C. x^2+8x+9

D. $x^2-8x+12$

356. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x+3)(x+4)$?

A. $x^2+7x+12$

B. $x^2+12x+7$

C. x^2+7x+8

D. $x^2-7x+12$

357. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+4)(x+7)$?
A. $x^2+11x+12$
B. $x^2-11x+28$
C. $x^2+11x+28$
D. $x^2+28x+11$

358. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+6)(x+1)$?
A. x^2+6x+7
B. x^2+7x+8
C. x^2-7x+6
D. x^2+7x+6

359. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+4)(x+3)$?
A. $x^2+7x+12$
B. $x^2+12x+7$
C. x^2+7x+8
D. $x^2-7x+12$

360. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+2)(x+5)$?
A. $x^2+7x+10$
B. $x^2+10x+7$
C. x^2+7x+8
D. $x^2-7x+10$

361. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+1)(x+5)$?
A. x^2+6x+5
B. x^2+5x+6
C. x^2+6x+7
D. x^2-6x+5

362. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+7)(x+2)$?
A. $x^2+9x+10$
B. $x^2-9x+14$
C. $x^2+9x+14$
D. $x^2+14x+9$

363. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+1)(x+6)$?
A. x^2+7x+8
B. x^2-7x+6
C. x^2+7x+6
D. x^2+6x+7

364. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+1)(x+3)$?
A. $x^2 - 4x + 3$
B. $x^2 + 4x + 3$
C. $x^2 + 3x + 4$
D. $x^2 + 4x + 5$

365. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+7)(x+5)$?
A. $x^2 + 12x + 13$
B. $x^2 - 12x + 35$
C. $x^2 + 12x + 35$
D. $x^2 + 35x + 12$

366. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+8)(x+4)$?
A. $x^2 + 32x + 12$
B. $x^2 + 12x + 13$
C. $x^2 - 12x + 32$
D. $x^2 + 12x + 32$

367. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+5)(x+2)$?
A. $x^2 + 7x + 8$
B. $x^2 - 7x + 10$
C. $x^2 + 7x + 10$
D. $x^2 + 10x + 7$

368. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+4)(x+2)$?
A. $x^2 - 6x + 8$
B. $x^2 + 6x + 8$
C. $x^2 + 8x + 6$
D. $x^2 + 6x + 7$

369. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+5)(x+1)$?
A. $x^2 - 6x + 5$
B. $x^2 + 6x + 5$
C. $x^2 + 5x + 6$
D. $x^2 + 6x + 7$

370. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+4)(x+4)$?
A. $x^2 - 8x + 16$
B. $x^2 + 8x + 16$
C. $x^2 + 16x + 8$
D. $x^2 + 8x + 9$

371. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+8)(x+7)$?
A. $x^2+15x+16$
B. $x^2-15x+56$
C. $x^2+15x+56$
D. $x^2+56x+15$

372. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+4)(x+8)$?
A. $x^2-12x+32$
B. $x^2+12x+32$
C. $x^2+32x+12$
D. $x^2+12x+13$

373. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+2)(x+8)$?
A. $x^2-10x+16$
B. $x^2+10x+16$
C. $x^2+16x+10$
D. $x^2+10x+11$

374. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+2)(x+7)$?
A. $x^2+9x+14$
B. $x^2+14x+9$
C. $x^2+9x+10$
D. $x^2-9x+14$

375. Equivalent expressions · Easy · MC
Which expression is equivalent to $(x+6)(x+7)$?
A. $x^2+13x+42$
B. $x^2+42x+13$
C. $x^2+13x+14$
D. $x^2-13x+42$

376. Equivalent expressions · Medium · MC
Which expression is equivalent to x^2+6x+5?
A. $(x-1)(x-5)$
B. $(x+1)(x+5)$
C. $(x+6)(x+5)$
D. $(x+1)(x-5)$

377. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2+9x+18$?
A. $(x+3)(x-6)$
B. $(x-3)(x-6)$
C. $(x+3)(x+6)$
D. $(x+9)(x+18)$

378. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 7x + 6$?
A. $(x + 1)(x + 6)$
B. $(x + 7)(x + 6)$
C. $(x + 1)(x - 6)$
D. $(x - 1)(x - 6)$

379. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 9x + 20$?
A. $(x + 5)(x + 4)$
B. $(x + 9)(x + 20)$
C. $(x + 5)(x - 4)$
D. $(x - 5)(x - 4)$

380. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 10x + 16$?
A. $(x - 2)(x - 8)$
B. $(x + 2)(x + 8)$
C. $(x + 10)(x + 16)$
D. $(x + 2)(x - 8)$

381. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 14x + 49$?
A. $(x + 14)(x + 49)$
B. $(x + 7)(x - 7)$
C. $(x - 7)(x - 7)$
D. $(x + 7)(x + 7)$

382. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 4x + 4$?
A. $(x + 2)(x - 2)$
B. $(x - 2)(x - 2)$
C. $(x + 2)(x + 2)$
D. $(x + 4)(x + 4)$

383. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 10x + 21$?
A. $(x + 10)(x + 21)$
B. $(x + 7)(x - 3)$
C. $(x - 7)(x - 3)$
D. $(x + 7)(x + 3)$

384. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 3x + 2$?
A. $(x - 1)(x - 2)$
B. $(x + 1)(x + 2)$
C. $(x + 3)(x + 2)$
D. $(x + 1)(x - 2)$

385. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 16x + 64$?
A. $(x - 8)(x - 8)$
B. $(x + 8)(x + 8)$
C. $(x + 16)(x + 64)$
D. $(x + 8)(x - 8)$

386. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 6x + 8$?
A. $(x + 6)(x + 8)$
B. $(x + 2)(x - 4)$
C. $(x - 2)(x - 4)$
D. $(x + 2)(x + 4)$

387. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 8x + 7$?
A. $(x + 7)(x - 1)$
B. $(x - 7)(x - 1)$
C. $(x + 7)(x + 1)$
D. $(x + 8)(x + 7)$

388. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 15x + 56$?
A. $(x - 8)(x - 7)$
B. $(x + 8)(x + 7)$
C. $(x + 15)(x + 56)$
D. $(x + 8)(x - 7)$

389. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 7x + 10$?
A. $(x + 7)(x + 10)$
B. $(x + 2)(x - 5)$
C. $(x - 2)(x - 5)$
D. $(x + 2)(x + 5)$

390. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 4x + 3$?
A. $(x + 3)(x + 1)$
B. $(x + 4)(x + 3)$
C. $(x + 3)(x - 1)$
D. $(x - 3)(x - 1)$

391. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 12x + 32$?
A. $(x + 12)(x + 32)$
B. $(x + 4)(x - 8)$
C. $(x - 4)(x - 8)$
D. $(x + 4)(x + 8)$

392. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 11x + 24$?

A. $(x + 3)(x + 8)$

B. $(x + 11)(x + 24)$

C. $(x + 3)(x - 8)$

D. $(x - 3)(x - 8)$

393. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 5x + 6$?

A. $(x + 3)(x - 2)$

B. $(x - 3)(x - 2)$

C. $(x + 3)(x + 2)$

D. $(x + 5)(x + 6)$

394. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 12x + 35$?

A. $(x - 5)(x - 7)$

B. $(x + 5)(x + 7)$

C. $(x + 12)(x + 35)$

D. $(x + 5)(x - 7)$

395. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 7x + 12$?

A. $(x - 3)(x - 4)$

B. $(x + 3)(x + 4)$

C. $(x + 7)(x + 12)$

D. $(x + 3)(x - 4)$

396. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 13x + 42$?

A. $(x + 7)(x + 6)$

B. $(x + 13)(x + 42)$

C. $(x + 7)(x - 6)$

D. $(x - 7)(x - 6)$

397. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 9x + 14$?

A. $(x - 2)(x - 7)$

B. $(x + 2)(x + 7)$

C. $(x + 9)(x + 14)$

D. $(x + 2)(x - 7)$

398. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 8x + 12$?

A. $(x + 8)(x + 12)$

B. $(x + 2)(x - 6)$

C. $(x - 2)(x - 6)$

D. $(x + 2)(x + 6)$

399. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 10x + 24$?
A. $(x - 4)(x - 6)$
B. $(x + 4)(x + 6)$
C. $(x + 10)(x + 24)$
D. $(x + 4)(x - 6)$

400. Equivalent expressions · Medium · MC
Which expression is equivalent to $x^2 + 9x + 8$?
A. $(x + 1)(x - 8)$
B. $(x - 1)(x - 8)$
C. $(x + 1)(x + 8)$
D. $(x + 9)(x + 8)$

401. Nonlinear equations · Easy · SPR
What is the sum of the solutions of $x^2 + 3x - 10 = 0$?

402. Nonlinear equations · Easy · SPR
What is the sum of the solutions of $x^2 - 6x - 7 = 0$?

403. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 + 1x = 0$?
A. 0
B. -2
C. 1
D. -1

404. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 10x + 24 = 0$?
A. 4
B. 24
C. 10
D. 6

405. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 5x + 4 = 0$?
A. 4
B. 5
C. 1
D. 3

406. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 4 = 0$?
A. -4
B. 0
C. 2
D. -2

407. Nonlinear equations · Easy · SPR
What is the sum of the solutions of $x^2 - 3x - 28 = 0$?

408. Nonlinear equations · Easy · SPR
What is the sum of the solutions of $x^2 - 11x + 28 = 0$?

409. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 6x + 5 = 0$?
A. 6
B. 4
C. 1
D. 5

410. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 + 7x + 10 = 0$?
A. -5
B. -2
C. -7
D. 10

411. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 + 5x + 4 = 0$?
A. -4
B. -5
C. 4
D. -1

412. Nonlinear equations · Easy · SPR
What is the sum of the solutions of $x^2 - 7x + 10 = 0$?

413. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 10x + 9 = 0$?
A. 9
B. 8
C. 10
D. 1

414. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 6x = 0$?
A. 6
B. 7
C. 5
D. 0

415. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 + 1x - 6 = 0$?
A. 2
B. -3
C. -1
D. -6

416. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 4x - 45 = 0$?
A. -5
B. 9
C. -45
D. 4

417. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 16 = 0$?
A. 4
B. -4
C. 0
D. -16

418. Nonlinear equations · Easy · SPR
What is the sum of the solutions of $x^2 - 1x - 12 = 0$?

419. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 9x + 20 = 0$?
A. 4
B. 20
C. 9
D. 5

420. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 + 2x - 8 = 0$?
A. -2
B. 2
C. -8
D. -4

421. Nonlinear equations · Easy · SPR
What is the sum of the solutions of $x^2 + 4x = 0$?

422. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 + 6x + 5 = 0$?
A. -5
B. 5
C. -6
D. -1

423. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 5x - 6 = 0$?
A. -6
B. -1
C. 6
D. 5

424. Nonlinear equations · Easy · SPR
What is the sum of the solutions of $x^2 - 9x = 0$?

425. Nonlinear equations · Easy · MC
What is the sum of the solutions of $x^2 - 3x - 4 = 0$?
A. 3
B. 4
C. -1
D. -4

426. Nonlinear equations · Medium · SPR
What is the greater solution of $x^2 - 15x + 54 = 0$?

427. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 11x + 24 = 0$?
A. 11
B. 24
C. 3
D. 8

428. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 7x + 10 = 0$?
A. 2
B. 7
C. 5
D. 10

429. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 + 1x - 20 = 0$?
A. -5
B. -1
C. 4
D. -20

430. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 2x - 15 = 0$?
A. 5
B. 2
C. -3
D. -15

431. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 + 5x + 6 = 0$?
A. -2
B. -3
C. 6
D. -5

432. Nonlinear equations · Medium · SPR
What is the greater solution of $x^2 - 7x + 12 = 0$?

433. Nonlinear equations · Medium · SPR
What is the greater solution of $x^2 + 7x + 10 = 0$?

434. Nonlinear equations · Medium · SPR
What is the greater solution of $x^2 - 1x - 6 = 0$?

435. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 2x - 8 = 0$?
A. -8
B. 2
C. 4
D. -2

436. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 6x = 0$?
A. 6
B. 5
C. 7
D. 0

437. Nonlinear equations · Medium · SPR
What is the greater solution of $x^2 - 15x + 56 = 0$?

438. Nonlinear equations · Medium · SPR
What is the greater solution of $x^2 - 9x + 14 = 0$?

439. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 10x + 21 = 0$?
A. 3
B. 10
C. 21
D. 7

440. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 5x - 6 = 0$?
A. -1
B. 5
C. -6
D. 6

441. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 + 4x - 5 = 0$?
A. 2
B. 1
C. -5
D. -4

442. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 + 2x - 3 = 0$?
A. -3
B. 1
C. -2
D. 2

443. Nonlinear equations · Medium · SPR
What is the greater solution of $x^2 - 13x + 40 = 0$?

444. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 2x = 0$?
A. 1
B. 0
C. 2
D. 3

445. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 5x - 14 = 0$?
A. -14
B. 5
C. -2
D. 7

446. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 7x - 8 = 0$?
A. 7
B. -8
C. -1
D. 8

447. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 3x + 2 = 0$?
A. 2
B. 3
C. 1
D. 4

448. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 1x - 30 = 0$?
A. 6
B. -5
C. -30
D. 1

449. Nonlinear equations · Medium · MC
What is the greater solution of $x^2 - 2x - 35 = 0$?
A. 7
B. -5
C. 2
D. -35

450. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 6x + 8 = 0$?

A. 8

B. 6

C. 4

D. 2

451. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 10x + 32$ is defined for all real x. What is the minimum value of f?

A. -5

B. 7

C. 5

D. 32

452. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 2x + -3$ is defined for all real x. What is the minimum value of f?

A. -1

B. 1

C. -4

D. -3

453. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 8x + 10$ is defined for all real x. What is the minimum value of f?

A. 4

B. 10

C. -4

D. -6

454. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 14x + 49$ is defined for all real x. What is the minimum value of f?

A. -7

B. 0

C. 7

D. 49

455. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 4x + 10$ is defined for all real x. What is the minimum value of f?

A. 6

B. 2

C. 10

D. -2

456. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 4x + 12$ is defined for all real x. What is the minimum value of f?

A. 8

B. -2

C. 12

D. 2

457. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 16x + 59$ is defined for all real x. What is the minimum value of f?

A. 8

B. 59

C. -8

D. -5

458. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 2x + -2$ is defined for all real x. What is the minimum value of f?

A. -1

B. -2

C. 1

D. -3

459. Nonlinear functions · Hard · SPR

The function $f(x) = x^2 - 2x + 8$ is defined for all real x. What is the minimum value of f?

460. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 14x + 46$ is defined for all real x. What is the minimum value of f?

A. 46

B. -3

C. -7

D. 7

461. Nonlinear functions · Hard · SPR

The function $f(x) = x^2 - 4x + 13$ is defined for all real x. What is the minimum value of f?

462. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 8x + 22$ is defined for all real x. What is the minimum value of f?

A. -4

B. 4

C. 22

D. 6

463. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 12x + 32$ is defined for all real x. What is the minimum value of f?

A. 32

B. -4

C. 6

D. -6

464. Nonlinear functions · Hard · SPR

The function $f(x) = x^2 - 16x + 61$ is defined for all real x. What is the minimum value of f?

465. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 16x + 66$ is defined for all real x. What is the minimum value of f?

A. 2

B. 66

C. 8

D. -8

466. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 8x + 23$ is defined for all real x. What is the minimum value of f?

A. 7

B. 23

C. -4

D. 4

467. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 8x + 20$ is defined for all real x. What is the minimum value of f?

A. 8

B. 20

C. 4

D. -4

468. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 14x + 56$ is defined for all real x. What is the minimum value of f?

A. 7

B. -7

C. 14

D. 56

469. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 6x + 4$ is defined for all real x. What is the minimum value of f?

A. 4

B. -3

C. -5

D. 3

470. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 6x + 8$ is defined for all real x. What is the minimum value of f?

A. -3

B. 8

C. -1

D. 3

471. Nonlinear functions · Easy · MC

If $f(t) = 2 \cdot 3^t$, what is the value of $f(2)$?

A. 18

B. 11

C. 12

D. 9

472. Nonlinear functions · Easy · SPR

If $f(t) = 8 \cdot 4^t$, what is the value of $f(4)$?

473. Nonlinear functions · Easy · SPR

If $f(t) = 8 \cdot 2^t$, what is the value of $f(3)$?

474. Nonlinear functions · Easy · MC
If $f(t) = 3 \cdot 3^t$, what is the value of $f(2)$?
A. 9
B. 12
C. 27
D. 18

475. Nonlinear functions · Easy · MC
If $f(t) = 10 \cdot 3^t$, what is the value of $f(3)$?
A. 270
B. 27
C. 37
D. 90

476. Nonlinear functions · Easy · SPR
If $f(t) = 11 \cdot 2^t$, what is the value of $f(2)$?

477. Nonlinear functions · Easy · SPR
If $f(t) = 4 \cdot 3^t$, what is the value of $f(4)$?

478. Nonlinear functions · Easy · MC
If $f(t) = 9 \cdot 2^t$, what is the value of $f(3)$?
A. 8
B. 54
C. 72
D. 17

479. Nonlinear functions · Easy · SPR
If $f(t) = 10 \cdot 4^t$, what is the value of $f(3)$?

480. Nonlinear functions · Easy · SPR
If $f(t) = 7 \cdot 3^t$, what is the value of $f(2)$?

481. Nonlinear functions · Easy · MC
If $f(t) = 8 \cdot 2^t$, what is the value of $f(2)$?
A. 12
B. 32
C. 4
D. 18

482. Nonlinear functions · Easy · MC
If $f(t) = 12 \cdot 4^t$, what is the value of $f(4)$?
A. 192
B. 256
C. 3072
D. 268

483. Nonlinear functions · Easy · MC
If $f(t) = 5 \cdot 3^t$, what is the value of $f(4)$?
A. 81
B. 86
C. 60
D. 405

484. Nonlinear functions · Easy · MC
If $f(t) = 8 \cdot 2^t$, what is the value of $f(4)$?
A. 64
B. 128
C. 16
D. 24

485. Nonlinear functions · Easy · SPR
If $f(t) = 2 \cdot 2^t$, what is the value of $f(3)$?

486. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 200 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?
A. 1200
B. 9
C. 1800
D. 206

487. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 20 \cdot 3^t$, where t is the number of years after the count begins. What is $P(3)$?
A. 29
B. 27
C. 180
D. 540

488. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 40 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?
A. 46
B. 240
C. 9
D. 360

489. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 50 \cdot 2^t$, where t is the number of years after the count begins. What is $P(4)$?
A. 800
B. 16
C. 58
D. 400

490. Nonlinear functions · Medium · SPR
A population is modeled by $P(t) = 125 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?

491. Nonlinear functions · Medium · SPR
A population is modeled by $P(t) = 100 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?

492. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 250 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?
A. 9
B. 1500
C. 256
D. 2250

493. Nonlinear functions · Medium · SPR
A population is modeled by $P(t) = 80 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?

494. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 250 \cdot 3^t$, where t is the number of years after the count begins. What is $P(4)$?
A. 262
B. 3000
C. 81
D. 20250

495. Nonlinear functions · Medium · SPR
A population is modeled by $P(t) = 200 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?

496. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 80 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?
A. 9
B. 86
C. 480
D. 720

497. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 125 \cdot 3^t$, where t is the number of years after the count begins. What is $P(4)$?
A. 10125
B. 1500
C. 81
D. 137

498. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 20 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?
A. 120
B. 180
C. 26
D. 9

499. Nonlinear functions · Medium · MC
A population is modeled by $P(t) = 250 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?
A. 8
B. 1500
C. 256
D. 2000

500. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 125 \cdot 2^t$, where t is the number of years after the count begins. What is $P(4)$?

A. 133

B. 16

C. 2000

D. 1000

501. Nonlinear functions · Hard · SPR

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 162$, what is the value of a?

502. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 112$, what is the value of a?

A. 56

B. 16

C. 7

D. 28

503. Nonlinear functions · Hard · SPR

A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 192$, what is the value of a?

504. Nonlinear functions · Hard · SPR

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 27$, what is the value of a?

505. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 2^t$. If $P(2) = 72$, what is the value of a?

A. 4

B. 72

C. 18

D. 36

506. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 171$, what is the value of a?

A. 9

B. 114

C. 57

D. 19

507. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 2^t$. If $P(2) = 32$, what is the value of a?

A. 16

B. 4

C. 32

D. 8

508. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 2^t$. If $P(2) = 28$, what is the value of a?

A. 14

B. 4

C. 28

D. 7

509. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 288$, what is the value of a?

A. 72

B. 16

C. 144

D. 18

510. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 90$, what is the value of a?

A. 30

B. 9

C. 10

D. 60

511. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^2 \cdot x^5$ for $x \neq 0$?

A. x^3

B. $10x$

C. x^7

D. x^{10}

512. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^2 \cdot x^4$ for $x \neq 0$?

A. x^2

B. $8x$

C. x^6

D. x^8

513. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^4 \cdot x^3$ for $x \neq 0$?

A. x^7

B. x^{12}

C. x^1

D. $12x$

514. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^2 \cdot x^3$ for $x \neq 0$?

A. x^5

B. x^6

C. x^1

D. $6x$

515. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^3 \cdot x^5$ for $x \neq 0$?

A. $15x$

B. x^8

C. x^{15}

D. x^2

516. Equivalent expressions · Easy · MC
Which expression is equivalent to $x^4 \cdot x^5$ for $x \neq 0$?
A. x^{20}
B. x^1
C. $20x$
D. x^9

517. Equivalent expressions · Easy · MC
Which expression is equivalent to $x^4 \cdot x^2$ for $x \neq 0$?
A. x^2
B. $8x$
C. x^6
D. x^8

518. Equivalent expressions · Easy · MC
Which expression is equivalent to $x^5 \cdot x^2$ for $x \neq 0$?
A. $10x$
B. x^7
C. x^{10}
D. x^3

519. Equivalent expressions · Easy · MC
Which expression is equivalent to $x^5 \cdot x^3$ for $x \neq 0$?
A. $15x$
B. x^8
C. x^{15}
D. x^2

520. Equivalent expressions · Easy · MC
Which expression is equivalent to $x^6 \cdot x^5$ for $x \neq 0$?
A. x^{30}
B. x^1
C. $30x$
D. x^{11}

521. Equivalent expressions · Easy · MC
Which expression is equivalent to $x^5 \cdot x^5$ for $x \neq 0$?
A. x^{25}
B. x^0
C. $25x$
D. x^{10}

522. Equivalent expressions · Easy · MC
Which expression is equivalent to $x^4 \cdot x^4$ for $x \neq 0$?
A. x^8
B. x^{16}
C. x^0
D. $16x$

523. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^4)^5$?
A. x^4
B. $20x$
C. x^{20}
D. x^9

524. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^5)^4$?
A. $20x$
B. x^{20}
C. x^9
D. x^5

525. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^4)^4$?
A. x^8
B. x^4
C. $16x$
D. x^{16}

526. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^6)^3$?
A. x^9
B. x^6
C. $18x$
D. x^{18}

527. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^3)^4$?
A. $12x$
B. x^{12}
C. x^7
D. x^3

528. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^3)^5$?
A. x^3
B. $15x$
C. x^{15}
D. x^8

529. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^2)^3$?
A. x^5
B. x^2
C. $6x$
D. x^6

530. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^3)^2$?
A. x^6
B. x^5
C. x^3
D. $6x$

531. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^5)^5$?
A. x^{25}
B. x^{10}
C. x^5
D. $25x$

532. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^4)^3$?
A. x^{12}
B. x^7
C. x^4
D. $12x$

533. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^6)^4$?
A. x^{10}
B. x^6
C. $24x$
D. x^{24}

534. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^2)^4$?
A. x^2
B. $8x$
C. x^8
D. x^6

535. Equivalent expressions · Medium · MC
Which expression is equivalent to $(x^2)^5$?
A. $10x$
B. x^{10}
C. x^7
D. x^2

536. Nonlinear functions · Easy · SPR
If $f(x) = 4x + 3$, what is the value of $f(-3)$?

537. Nonlinear functions · Easy · SPR
If $f(x) = 6x - 5$, what is the value of $f(4)$?

538. Nonlinear functions · Easy · MC
If $f(x) = 6x + 3$, what is the value of $f(2)$?
A. 15
B. 9
C. 11
D. 30

539. Nonlinear functions · Easy · SPR
If $f(x) = 6x - 4$, what is the value of $f(1)$?

540. Nonlinear functions · Easy · SPR
If $f(x) = 3x + 6$, what is the value of $f(-1)$?

541. Nonlinear functions · Easy · MC
If $f(x) = 2x - 4$, what is the value of $f(-3)$?
A. -10
B. -5
C. -14
D. -2

542. Nonlinear functions · Easy · MC
If $f(x) = 5x + 6$, what is the value of $f(3)$?
A. 45
B. 14
C. 21
D. 9

543. Nonlinear functions · Easy · SPR
If $f(x) = 4x + 5$, what is the value of $f(3)$?

544. Nonlinear functions · Easy · SPR
If $f(x) = 6x$, what is the value of $f(0)$?

545. Nonlinear functions · Easy · MC
If $f(x) = 3x + 1$, what is the value of $f(-1)$?
A. 0
B. -4
C. 3
D. -2

546. Nonlinear functions · Easy · SPR
If $f(x) = 2x + 4$, what is the value of $f(6)$?

547. Nonlinear functions · Easy · MC
If $f(x) = 6x - 4$, what is the value of $f(-3)$?
A. -22
B. -42
C. -14
D. -1

548. Nonlinear functions · Easy · SPR
If $f(x) = 3x + 7$, what is the value of $f(-1)$?

549. Nonlinear functions · Easy · MC
If $f(x) = 4x + 5$, what is the value of $f(0)$?
A. 20
B. 9
C. 5
D. -5

550. Nonlinear functions · Easy · MC
If $f(x) = 2x - 3$, what is the value of $f(2)$?
A. 7
B. 1
C. -2
D. 2

551. Nonlinear functions · Easy · MC
If $f(x) = 2x - 5$, what is the value of $f(-3)$?
A. -16
B. -11
C. -6
D. -1

552. Nonlinear functions · Easy · MC
If $f(x) = 4x - 2$, what is the value of $f(-3)$?
A. -1
B. -20
C. -14
D. -10

553. Nonlinear functions · Easy · MC
If $f(x) = 5x - 3$, what is the value of $f(6)$?
A. 15
B. 8
C. 33
D. 27

554. Nonlinear functions · Easy · SPR
If $f(x) = 4x + 7$, what is the value of $f(6)$?

555. Nonlinear functions · Easy · SPR
If $f(x) = 6x$, what is the value of $f(7)$?

556. Nonlinear functions · Medium · MC
If $f(x) = 4x + 5$ and $g(x) = 2x + 3$, what is the value of $f(g(-1))$?
A. -8
B. 9
C. 2
D. 10

557. Nonlinear functions · Medium · MC
If $f(x) = 2x + 5$ and $g(x) = 2x$, what is the value of $f(g(2))$?
A. 8
B. 12
C. 13
D. 14

558. Nonlinear functions · Medium · MC
If $f(x) = 3x + 3$ and $g(x) = 4x + 1$, what is the value of $f(g(2))$?
A. 24
B. 18
C. 31
D. 30

559. Nonlinear functions · Medium · MC
If $f(x) = 3x + 1$ and $g(x) = 3x - 1$, what is the value of $f(g(-2))$?
A. -20
B. -19
C. -18
D. -12

560. Nonlinear functions · Medium · MC
If $f(x) = 3x - 4$ and $g(x) = 4x - 1$, what is the value of $f(g(-2))$?
A. -31
B. -30
C. -19
D. -24

561. Nonlinear functions · Medium · MC
If $f(x) = 3x - 4$ and $g(x) = 4x + 4$, what is the value of $f(g(3))$?
A. 36
B. 21
C. 44
D. 45

562. Nonlinear functions · Medium · MC
If $f(x) = 4x - 1$ and $g(x) = 4x - 3$, what is the value of $f(g(4))$?
A. 28
B. 51
C. 64
D. 52

563. Nonlinear functions · Medium · MC
If $f(x) = 2x - 3$ and $g(x) = 4x + 5$, what is the value of $f(g(0))$?
A. 2
B. 0
C. 8
D. 7

564. Nonlinear functions · Medium · MC

If $f(x) = 5x - 1$ and $g(x) = 4x - 2$, what is the value of $f(g(-1))$?

A. -31

B. -20

C. -30

D. -12

565. Nonlinear functions · Medium · MC

If $f(x) = 2x + 1$ and $g(x) = 4x - 1$, what is the value of $f(g(5))$?

A. 39

B. 38

C. 30

D. 40

566. Nonlinear functions · Medium · MC

If $f(x) = 2x + 1$ and $g(x) = 4x + 1$, what is the value of $f(g(4))$?

A. 35

B. 26

C. 32

D. 36

567. Nonlinear functions · Medium · MC

If $f(x) = 4x + 6$ and $g(x) = 2x + 1$, what is the value of $f(g(5))$?

A. 40

B. 51

C. 50

D. 37

568. Nonlinear functions · Medium · MC

If $f(x) = 2x$ and $g(x) = 2x - 2$, what is the value of $f(g(-2))$?

A. -10

B. -8

C. -12

D. -11

569. Nonlinear functions · Medium · MC

If $f(x) = 3x + 3$ and $g(x) = 2x - 1$, what is the value of $f(g(1))$?

A. 5

B. 7

C. 6

D. 8

570. Nonlinear functions · Medium · MC

If $f(x) = 2x + 3$ and $g(x) = 3x + 2$, what is the value of $f(g(4))$?

A. 25

B. 31

C. 24

D. 32

571. Nonlinear functions · Medium · MC

If $f(x) = 3x - 3$ and $g(x) = 4x + 5$, what is the value of $f(g(4))$?

A. 60

B. 30

C. 61

D. 48

572. Nonlinear functions · Medium · MC

If $f(x) = 3x - 2$ and $g(x) = 2x + 3$, what is the value of $f(g(-1))$?

A. -4

B. 1

C. 2

D. -6

573. Nonlinear functions · Medium · MC

If $f(x) = 3x - 4$ and $g(x) = 3x + 5$, what is the value of $f(g(1))$?

A. 20

B. 7

C. 21

D. 9

574. Nonlinear functions · Medium · SPR

If $f(x) = 4x - 2$ and $g(x) = 3x + 3$, what is the value of $f(g(0))$?

575. Nonlinear functions · Medium · MC

If $f(x) = 3x + 4$ and $g(x) = 4x - 3$, what is the value of $f(g(3))$?

A. 36

B. 31

C. 32

D. 22

576. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 2)$. If $f(6) = -5$, what is the value of $g(8)$?

A. -7

B. -5

C. 6

D. -3

577. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 4)$. If $f(-1) = 9$, what is the value of $g(3)$?

A. -1

B. 5

C. 13

D. 9

578. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 1)$. If $f(4) = 1$, what is the value of $g(5)$?

A. 4

B. 0

C. 1

D. 2

579. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 3)$. If $f(3) = 6$, what is the value of $g(6)$?

A. 3

B. 9

C. 12

D. 6

580. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 2)$. If $f(4) = -1$, what is the value of $g(6)$?

A. 4

B. -1

C. 1

D. -3

581. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 3)$. If $f(2) = -1$, what is the value of $g(5)$?

A. -4

B. 4

C. 2

D. -1

582. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 1)$. If $f(2) = 5$, what is the value of $g(3)$?

A. 2

B. 6

C. 5

D. 4

583. Nonlinear functions · Hard · SPR

The function g is defined by $g(x) = f(x - 4)$. If $f(0) = 7$, what is the value of $g(4)$?

584. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 1)$. If $f(4) = 4$, what is the value of $g(5)$?

A. 3

B. 9

C. 4

D. 5

585. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 2)$. If $f(5) = 10$, what is the value of $g(7)$?

A. 5

B. 12

C. 10

D. 8

586. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 1)$. If $f(4) = -4$, what is the value of $g(5)$?

A. -3

B. -4

C. 4

D. -5

587. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 3)$. If $f(4) = -3$, what is the value of $g(7)$?

A. 0

B. -3

C. -6

D. 4

588. Nonlinear functions · Hard · SPR

The function g is defined by $g(x) = f(x - 2)$. If $f(4) = 8$, what is the value of $g(6)$?

589. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 2)$. If $f(6) = 0$, what is the value of $g(8)$?

A. 0

B. -2

C. 6

D. 2

590. Nonlinear functions · Hard · SPR

The function g is defined by $g(x) = f(x - 4)$. If $f(1) = 5$, what is the value of $g(5)$?

591. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x + 6} = 7$?

A. 49

B. 43

C. 55

D. 13

592. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x + 12} = 6$?

A. 24

B. 48

C. 18

D. 36

593. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+2} = 4$?
A. 16
B. 18
C. 6
D. 14

594. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+1} = 2$?
A. 3
B. 4
C. -1
D. 5

595. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+1} = 3$?
A. 8
B. 4
C. 9
D. 10

596. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+4} = 4$?
A. 8
B. 12
C. 20
D. 16

597. Nonlinear equations · Medium · SPR
What is the solution of $\sqrt{x+5} = 9$?

598. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+8} = 3$?
A. 9
B. 17
C. 11
D. 1

599. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+1} = 9$?
A. 10
B. 81
C. 82
D. 80

600. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+6}=5$?
A. 19
B. 11
C. 31
D. 25

601. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+2}=6$?
A. 34
B. 8
C. 38
D. 36

602. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+7}=9$?
A. 81
B. 74
C. 88
D. 16

603. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+7}=7$?
A. 42
B. 14
C. 56
D. 49

604. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+2}=8$?
A. 62
B. 66
C. 64
D. 10

605. Nonlinear equations · Medium · MC
What is the solution of $\sqrt{x+5}=3$?
A. 9
B. 4
C. 8
D. 14

606. Nonlinear equations · Hard · MC
What is the solution of $\sqrt{x+11}=9$?
A. 70
B. 81
C. 92
D. 20

607. Nonlinear equations · Hard · MC
What is the solution of $\sqrt{x+4}=3$?
A. 13
B. 7
C. 5
D. 9

608. Nonlinear equations · Hard · MC
What is the solution of $\sqrt{x+3}=8$?
A. 64
B. 61
C. 11
D. 67

609. Nonlinear equations · Hard · SPR
What is the solution of $\sqrt{x+6}=6$?

610. Nonlinear equations · Hard · MC
What is the solution of $\sqrt{x+9}=7$?
A. 16
B. 40
C. 58
D. 49

611. Nonlinear equations · Hard · SPR
What is the solution of $\sqrt{x+11}=4$?

612. Nonlinear equations · Hard · MC
What is the solution of $\sqrt{x+3}=5$?
A. 8
B. 22
C. 25
D. 28

613. Nonlinear equations · Hard · SPR
What is the solution of $\sqrt{x+11}=5$?

614. Nonlinear equations · Hard · MC
What is the solution of $\sqrt{x+8}=9$?
A. 73
B. 17
C. 81
D. 89

615. Nonlinear equations · Hard · SPR
What is the solution of $\sqrt{x+5}=7$?

616. Nonlinear equations · Medium · SPR
What value of x satisfies $\dfrac{x+3}{x+2}=2$, given that the expression is defined?

617. Nonlinear equations · Medium · SPR

What value of x satisfies $\dfrac{x+4}{x+1} = 4$, given that the expression is defined?

618. Nonlinear equations · Medium · SPR

What value of x satisfies $\dfrac{x+8}{x+1} = 2$, given that the expression is defined?

619. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+9}{x+4} = 2$, given that the expression is defined?

A. 10

B. 3

C. 1

D. -4

620. Nonlinear equations · Medium · SPR

What value of x satisfies $\dfrac{x+20}{x+4} = 3$, given that the expression is defined?

621. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+22}{x+4} = 3$, given that the expression is defined?

A. 5

B. -4

C. 27

D. 8

622. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+28}{x+1} = 4$, given that the expression is defined?

A. -1

B. 8

C. 12

D. 36

623. Nonlinear equations · Medium · SPR

What value of x satisfies $\dfrac{x+17}{x+6} = 2$, given that the expression is defined?

624. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+37}{x+4} = 4$, given that the expression is defined?

A. 11

B. 44

C. -4

D. 7

625. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+15}{x+4} = 2$, given that the expression is defined?

A. -4

B. 22

C. 7

D. 9

626. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+18}{x+2} = 3$, given that the expression is defined?

A. -2

B. 9

C. 24

D. 6

627. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+13}{x+1} = 4$, given that the expression is defined?

A. 16

B. -1

C. 7

D. 3

628. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+4}{x+1} = 2$, given that the expression is defined?

A. 4

B. 6

C. -1

D. 2

629. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+14}{x+2} = 4$, given that the expression is defined?

A. -2

B. 6

C. 16

D. 2

630. Nonlinear equations · Medium · MC

What value of x satisfies $\dfrac{x+17}{x+1} = 3$, given that the expression is defined?

A. 10

B. 24

C. -1

D. 7

631. Nonlinear equations · Hard · SPR

What value of x satisfies $\dfrac{x+22}{x+6} = 3$, given that the expression is defined?

632. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x+35}{x+5} = 4$, given that the expression is defined?

A. 9

B. -5

C. 5

D. 40

633. Nonlinear equations · Hard · SPR

What value of x satisfies $\dfrac{x+32}{x+2}=4$, given that the expression is defined?

634. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x+14}{x+4}=3$, given that the expression is defined?

A. -4

B. 1

C. 15

D. 4

635. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x+29}{x+2}=4$, given that the expression is defined?

A. 11

B. 36

C. -2

D. 7

636. Nonlinear equations · Hard · SPR

What value of x satisfies $\dfrac{x+6}{x+2}=2$, given that the expression is defined?

637. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x+5}{x+3}=2$, given that the expression is defined?

A. 4

B. -3

C. 1

D. -1

638. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x+15}{x+5}=2$, given that the expression is defined?

A. 5

B. 7

C. 20

D. -5

639. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x+13}{x+4}=2$, given that the expression is defined?

A. -4

B. 18

C. 7

D. 5

640. Nonlinear equations · Hard · SPR

What value of x satisfies $\dfrac{x+18}{x+6}=2$, given that the expression is defined?

641. Nonlinear equations · Hard · SPR

What value of x satisfies $\dfrac{x+20}{x+6}=3$, given that the expression is defined?

642. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x+16}{x+4} = 3$, given that the expression is defined?

A. 18

B. 5

C. 2

D. -4

643. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x-1}{x+2} = 2$, given that the expression is defined?

A. -6

B. -2

C. -5

D. -3

644. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x+6}{x+5} = 2$, given that the expression is defined?

A. -4

B. 2

C. -5

D. -2

645. Nonlinear equations · Hard · MC

What value of x satisfies $\dfrac{x+11}{x+2} = 2$, given that the expression is defined?

A. -2

B. 18

C. 9

D. 7

646. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 11x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 12

B. 121

C. 0

D. 11

647. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 2x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 3

B. 4

C. 0

D. 2

648. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 16x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 0

B. 256

C. 16

D. 17

649. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 4x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 5

B. 16

C. 4

D. 0

650. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 13x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 169

B. 0

C. 13

D. 14

651. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 9x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 10

B. 9

C. 0

D. 81

652. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 7x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 0

B. 8

C. 49

D. 7

653. Systems of equations · Medium · SPR

The graphs of $y = x^2$ and $y = 3x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

654. Systems of equations · Medium · SPR

The graphs of $y = x^2$ and $y = 6x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

655. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 5x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 5

B. 0

C. 25

D. 6

656. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 8x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 64

B. 8

C. 0

D. 9

657. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 10x$ intersect at the origin and one other point. What is the positive x-coordinate of that other point?

A. 100

B. 11

C. 10

D. 0

658. Systems of equations · Hard · SPR

The graphs of $y = x^2$ and $y = 9x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

659. Systems of equations · Hard · SPR

The graphs of $y = x^2$ and $y = 11x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

660. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 6x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 36

B. 12

C. 216

D. 6

661. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 2x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 2

B. 0

C. 8

D. 4

662. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 13x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 169

B. 26

C. 13

D. 2197

663. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 10x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 20

B. 100

C. 1000

D. 10

664. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 8x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 16

B. 512

C. 64

D. 8

665. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 16x$ intersect at the origin and one other point. What is the y-coordinate of that other point?

A. 32

B. 4096

C. 16

D. 256

666. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 - 24x + 135 = 0$ have?

A. Exactly two

B. Infinitely many

C. None

D. Exactly one

667. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 4x + 4 = 0$ have?

A. Exactly two

B. Infinitely many

C. None

D. Exactly one

668. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 - 13x + 40 = 0$ have?
A. Exactly one
B. Exactly two
C. Infinitely many
D. None

669. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 6x + 15 = 0$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

670. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 - 22x + 117 = 0$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

671. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 6x + 17 = 0$ have?
A. Exactly two
B. Infinitely many
C. None
D. Exactly one

672. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 8x + 21 = 0$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

673. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 2x + 11 = 0$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

674. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 14x + 49 = 0$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

675. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 - 4x + 4 = 0$ have?
A. Infinitely many
B. None
C. Exactly one
D. Exactly two

676. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 8 = 0$ have?
A. Exactly one
B. Exactly two
C. Infinitely many
D. None

677. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 6x + 9 = 0$ have?
A. Infinitely many
B. None
C. Exactly one
D. Exactly two

678. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 1x + 8 = 0$ have?
A. Exactly one
B. Exactly two
C. Infinitely many
D. None

679. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 3x + 4 = 0$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

680. Nonlinear equations · Medium · MC
How many distinct real solutions does $x^2 + 7x + 16 = 0$ have?
A. None
B. Exactly one
C. Exactly two
D. Infinitely many

681. Equivalent expressions · Hard · MC
Which expression is equivalent to $(x - 4)(x + 5) - 2(x - 4)$?
A. $(x - 4)(x + 3)$
B. $(x - 4)(x + 7)$
C. $(x + 4)(x + 3)$
D. $(x - 4)(x - 3)$

682. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-3)(x+4)-4(x-3)$?

A. $(x-3)(x+0)$

B. $(x-3)(x+8)$

C. $(x+3)(x+0)$

D. $(x-3)(x-0)$

683. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-3)(x+3)-3(x-3)$?

A. $(x-3)(x+0)$

B. $(x-3)(x+6)$

C. $(x+3)(x+0)$

D. $(x-3)(x-0)$

684. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-6)(x+7)-5(x-6)$?

A. $(x-6)(x+2)$

B. $(x-6)(x+12)$

C. $(x+6)(x+2)$

D. $(x-6)(x-2)$

685. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-2)(x+2)-4(x-2)$?

A. $(x+2)(x+-2)$

B. $(x-2)(x--2)$

C. $(x-2)(x+-2)$

D. $(x-2)(x+6)$

686. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-5)(x+3)-3(x-5)$?

A. $(x-5)(x+0)$

B. $(x-5)(x+6)$

C. $(x+5)(x+0)$

D. $(x-5)(x-0)$

687. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-2)(x+6)-2(x-2)$?

A. $(x+2)(x+4)$

B. $(x-2)(x-4)$

C. $(x-2)(x+4)$

D. $(x-2)(x+8)$

688. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-5)(x+7)-4(x-5)$?

A. $(x-5)(x+11)$

B. $(x+5)(x+3)$

C. $(x-5)(x-3)$

D. $(x-5)(x+3)$

689. Equivalent expressions · Hard · MC
Which expression is equivalent to $(x-6)(x+6)-3(x-6)$?
A. $(x-6)(x-3)$
B. $(x-6)(x+3)$
C. $(x-6)(x+9)$
D. $(x+6)(x+3)$

690. Equivalent expressions · Hard · MC
Which expression is equivalent to $(x-2)(x+5)-3(x-2)$?
A. $(x-2)(x-2)$
B. $(x-2)(x+2)$
C. $(x-2)(x+8)$
D. $(x+2)(x+2)$

691. Equivalent expressions · Hard · MC
Which expression is equivalent to $(x-2)(x+4)-4(x-2)$?
A. $(x+2)(x+0)$
B. $(x-2)(x-0)$
C. $(x-2)(x+0)$
D. $(x-2)(x+8)$

692. Equivalent expressions · Hard · MC
Which expression is equivalent to $(x-4)(x+6)-5(x-4)$?
A. $(x+4)(x+1)$
B. $(x-4)(x-1)$
C. $(x-4)(x+1)$
D. $(x-4)(x+11)$

693. Equivalent expressions · Hard · MC
Which expression is equivalent to $(x-5)(x+4)-5(x-5)$?
A. $(x+5)(x+-1)$
B. $(x-5)(x--1)$
C. $(x-5)(x+-1)$
D. $(x-5)(x+9)$

694. Equivalent expressions · Hard · MC
Which expression is equivalent to $(x-6)(x+2)-2(x-6)$?
A. $(x-6)(x+4)$
B. $(x+6)(x+0)$
C. $(x-6)(x-0)$
D. $(x-6)(x+0)$

695. Equivalent expressions · Hard · MC
Which expression is equivalent to $(x-6)(x+7)-3(x-6)$?
A. $(x-6)(x+10)$
B. $(x+6)(x+4)$
C. $(x-6)(x-4)$
D. $(x-6)(x+4)$

696. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-3)(x+7)-2(x-3)$?

A. $(x-3)(x+9)$

B. $(x+3)(x+5)$

C. $(x-3)(x-5)$

D. $(x-3)(x+5)$

697. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-3)(x+4)-3(x-3)$?

A. $(x-3)(x+1)$

B. $(x-3)(x+7)$

C. $(x+3)(x+1)$

D. $(x-3)(x-1)$

698. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-5)(x+4)-4(x-5)$?

A. $(x-5)(x-0)$

B. $(x-5)(x+0)$

C. $(x-5)(x+8)$

D. $(x+5)(x+0)$

699. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-4)(x+2)-3(x-4)$?

A. $(x-4)(x+5)$

B. $(x+4)(x+-1)$

C. $(x-4)(x--1)$

D. $(x-4)(x+-1)$

700. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x-2)(x+7)-5(x-2)$?

A. $(x-2)(x-2)$

B. $(x-2)(x+2)$

C. $(x-2)(x+12)$

D. $(x+2)(x+2)$

Problem-Solving and Data Analysis

701. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 2. There are 12 red marbles. How many marbles are in the bag?

A. 24

B. 60

C. 14

D. 20

702. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 3. There are 6 red marbles. How many marbles are in the bag?

A. 9

B. 36

C. 12

D. 18

703. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 2. There are 6 red marbles. How many marbles are in the bag?

A. 30

B. 12

C. 8

D. 10

704. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 2 : 3. There are 6 red marbles. How many marbles are in the bag?

A. 15

B. 18

C. 30

D. 9

705. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 4 : 6. There are 16 red marbles. How many marbles are in the bag?

A. 40

B. 96

C. 22

D. 160

706. Ratios and rates · Easy · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 5. There are 16 red marbles. How many marbles are in the bag?

707. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 5. There are 12 red marbles. How many marbles are in the bag?

A. 60

B. 96

C. 32

D. 17

708. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 2 : 6. There are 4 red marbles. How many marbles are in the bag?

A. 10

B. 32

C. 24

D. 16

709. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 4 : 2. There are 12 red marbles. How many marbles are in the bag?

A. 18

B. 14

C. 24

D. 72

710. Ratios and rates · Easy · SPR

The ratio of red marbles to blue marbles in a bag is 5 : 6. There are 20 red marbles. How many marbles are in the bag?

711. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 3 : 3. There are 18 red marbles. How many marbles are in the bag?

712. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 3. There are 20 red marbles. How many marbles are in the bag?

713. Ratios and rates · Medium · MC

The ratio of red marbles to blue marbles in a bag is 3 : 2. There are 18 red marbles. How many marbles are in the bag?

A. 36

B. 90

C. 30

D. 20

714. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 6. There are 12 red marbles. How many marbles are in the bag?

715. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 5. There are 24 red marbles. How many marbles are in the bag?

716. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 2. There are 24 red marbles. How many marbles are in the bag?

717. Ratios and rates · Medium · MC

The ratio of red marbles to blue marbles in a bag is 5 : 2. There are 40 red marbles. How many marbles are in the bag?

A. 80

B. 280

C. 56

D. 42

718. Ratios and rates · Medium · MC

The ratio of red marbles to blue marbles in a bag is 4 : 3. There are 24 red marbles. How many marbles are in the bag?

A. 168

B. 27

C. 72

D. 42

719. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 5 : 5. There are 25 red marbles. How many marbles are in the bag?

720. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 2 : 6. There are 8 red marbles. How many marbles are in the bag?

721. Ratios and rates · Hard · MC

The ratio of red marbles to blue marbles in a bag is 3 : 3. There are 9 red marbles. How many marbles are in the bag?

A. 54

B. 27

C. 12

D. 18

722. Ratios and rates · Hard · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 2. There are 16 red marbles. How many marbles are in the bag?

723. Ratios and rates · Hard · SPR

The ratio of red marbles to blue marbles in a bag is 5 : 3. There are 40 red marbles. How many marbles are in the bag?

724. Ratios and rates · Hard · MC

The ratio of red marbles to blue marbles in a bag is 4 : 5. There are 20 red marbles. How many marbles are in the bag?

A. 180

B. 100

C. 45

D. 25

725. Ratios and rates · Hard · MC

The ratio of red marbles to blue marbles in a bag is 5 : 3. There are 20 red marbles. How many marbles are in the bag?

A. 32

B. 160

C. 60

D. 23

726. Ratios and rates · Easy · MC

A car travels 210 miles in 5 hours at a constant speed. What is the speed in miles per hour?

A. 42

B. 5

C. 215

D. 1050

727. Ratios and rates · Easy · MC

A car travels 150 miles in 2 hours at a constant speed. What is the speed in miles per hour?

A. 152

B. 300

C. 75

D. 2

728. Ratios and rates · Easy · MC

A car travels 180 miles in 2 hours at a constant speed. What is the speed in miles per hour?

A. 2

B. 182

C. 360

D. 90

729. Ratios and rates · Easy · SPR

A car travels 300 miles in 2 hours at a constant speed. What is the speed in miles per hour?

730. Ratios and rates · Easy · MC

A car travels 240 miles in 2 hours at a constant speed. What is the speed in miles per hour?

A. 480

B. 2

C. 242

D. 120

731. Ratios and rates · Easy · SPR

A car travels 150 miles in 5 hours at a constant speed. What is the speed in miles per hour?

732. Ratios and rates · Easy · MC

A car travels 240 miles in 3 hours at a constant speed. What is the speed in miles per hour?

A. 3

B. 243

C. 80

D. 720

733. Ratios and rates · Easy · MC

A car travels 240 miles in 5 hours at a constant speed. What is the speed in miles per hour?

A. 1200

B. 5

C. 48

D. 245

734. Percentages · Easy · MC

A jacket costs $80. During a sale the price is reduced by 30%. What is the sale price, in dollars?

A. 104

B. 24

C. 2400

D. 56

735. Percentages · Easy · MC

A jacket costs $100. During a sale the price is reduced by 10%. What is the sale price, in dollars?

A. 1000

B. 90

C. 10

D. 110

736. Percentages · Easy · SPR

A jacket costs $200. During a sale the price is reduced by 20%. What is the sale price, in dollars?

737. Percentages · Easy · MC

A jacket costs $40. During a sale the price is reduced by 25%. What is the sale price, in dollars?

A. 30

B. 1000

C. 50

D. 10

738. Percentages · Easy · MC

A jacket costs $60. During a sale the price is reduced by 20%. What is the sale price, in dollars?

A. 12

B. 1200

C. 72

D. 48

739. Percentages · Easy · MC

A jacket costs $120. During a sale the price is reduced by 10%. What is the sale price, in dollars?

A. 12

B. 1200

C. 108

D. 132

740. Percentages · Easy · MC

A jacket costs \$50. During a sale the price is reduced by 15%. What is the sale price, in dollars?

A. $\frac{15}{2}$

B. $\frac{85}{2}$

C. $\frac{115}{2}$

D. 750

741. Percentages · Easy · MC

A jacket costs \$50. During a sale the price is reduced by 30%. What is the sale price, in dollars?

A. 35

B. 15

C. 1500

D. 65

742. Percentages · Easy · SPR

A jacket costs \$120. During a sale the price is reduced by 15%. What is the sale price, in dollars?

743. Percentages · Easy · MC

A jacket costs \$100. During a sale the price is reduced by 30%. What is the sale price, in dollars?

A. 3000

B. 130

C. 30

D. 70

744. Percentages · Easy · MC

A jacket costs \$60. During a sale the price is reduced by 15%. What is the sale price, in dollars?

A. 51

B. 900

C. 69

D. 9

745. Percentages · Easy · SPR

A jacket costs \$60. During a sale the price is reduced by 10%. What is the sale price, in dollars?

746. Percentages · Medium · MC

A bike is listed at \$100. It is discounted by 25%, and then a 8% sales tax is added to the discounted price. What is the final price, in dollars?

A. 81

B. 67

C. 83

D. 75

747. Percentages · Medium · SPR

A bike is listed at \$200. It is discounted by 25%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

748. Percentages · Medium · SPR

A bike is listed at \$50. It is discounted by 20%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

749. Percentages · Medium · MC

A bike is listed at $80. It is discounted by 25%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 68

B. 60

C. 66

D. 45

750. Percentages · Medium · MC

A bike is listed at $50. It is discounted by 20%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

A. 45

B. 25

C. 42

D. 40

751. Percentages · Medium · MC

A bike is listed at $200. It is discounted by 30%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 150

B. 140

C. 160

D. 154

752. Percentages · Medium · MC

A bike is listed at $200. It is discounted by 25%, and then a 8% sales tax is added to the discounted price. What is the final price, in dollars?

A. 167

B. 150

C. 162

D. 166

753. Percentages · Medium · SPR

A bike is listed at $200. It is discounted by 20%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

754. Percentages · Medium · MC

A bike is listed at $40. It is discounted by 25%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 34

B. 5

C. 30

D. 33

755. Percentages · Medium · MC

A bike is listed at $200. It is discounted by 10%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

A. 190

B. 180

C. 185

D. 189

756. Percentages · Hard · SPR

A bike is listed at $200. It is discounted by 30%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

757. Percentages · Hard · MC

A bike is listed at $120. It is discounted by 25%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 99

B. 102

C. 90

D. 85

758. Percentages · Hard · MC

A bike is listed at $100. It is discounted by 20%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

A. 85

B. 75

C. 80

D. 84

759. Percentages · Hard · MC

A bike is listed at $80. It is discounted by 25%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

A. 50

B. 63

C. 60

D. 64

760. Percentages · Hard · MC

A bike is listed at $200. It is discounted by 15%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 187

B. 170

C. 190

D. 175

761. Percentages · Hard · MC

A bike is listed at $100. It is discounted by 20%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 70

B. 80

C. 88

D. 90

762. Percentages · Hard · MC

A bike is listed at $100. It is discounted by 30%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 80

B. 77

C. 60

D. 70

763. Percentages · Hard · MC

A bike is listed at \$100. It is discounted by 10%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

A. 100

B. 80

C. 99

D. 90

764. One-variable data · Easy · MC

The mean of the numbers 6, 17, 6, 8, and x is 9. What is the value of x?

A. 45

B. 9

C. 7

D. 8

765. One-variable data · Easy · MC

The mean of the numbers 10, 12, 8, 11, and x is 12. What is the value of x?

A. 12

B. 20

C. 10

D. 19

766. One-variable data · Easy · MC

The mean of the numbers 19, 9, 11, 5, and x is 15. What is the value of x?

A. 11

B. 31

C. 32

D. 15

767. One-variable data · Easy · MC

The mean of the numbers 18, 4, 15, 10, and x is 10. What is the value of x?

A. 3

B. 11

C. 10

D. 4

768. One-variable data · Easy · SPR

The mean of the numbers 11, 9, 17, 18, and x is 14. What is the value of x?

769. One-variable data · Easy · SPR

The mean of the numbers 10, 20, 20, 14, and x is 15. What is the value of x?

770. One-variable data · Medium · MC

The mean of the numbers 12, 6, 16, 10, and x is 12. What is the value of x?

A. 16

B. 17

C. 11

D. 12

771. One-variable data · Medium · MC

The mean of the numbers 8, 10, 18, 8, and x is 16. What is the value of x?

A. 16

B. 11

C. 36

D. 37

772. One-variable data · Medium · MC

The mean of the numbers 8, 14, 5, 14, and x is 11. What is the value of x?

A. 11

B. 15

C. 10

D. 14

773. One-variable data · Medium · MC

The mean of the numbers 4, 6, 14, 17, and x is 16. What is the value of x?

A. 39

B. 40

C. 10

D. 16

774. One-variable data · Medium · MC

The mean of the numbers 16, 6, 14, 20, and x is 13. What is the value of x?

A. 9

B. 14

C. 13

D. 10

775. One-variable data · Medium · MC

The mean of the numbers 10, 12, 8, 11, and x is 10. What is the value of x?

A. 50

B. 8

C. 10

D. 9

776. One-variable data · Easy · MC

What is the median of the data set 5, 11, 16, 19, 26?

A. 15

B. 26

C. 5

D. 16

777. One-variable data · Easy · MC

What is the median of the data set 12, 17, 22, 24, 25?

A. 22

B. 20

C. 12

D. 25

778. One-variable data · Easy · MC
What is the median of the data set 6, 11, 13, 14, 26?
A. 14
B. 13
C. 6
D. 26

779. One-variable data · Easy · MC
What is the median of the data set 10, 12, 20, 24, 29?
A. 19
B. 29
C. 10
D. 20

780. One-variable data · Easy · MC
What is the median of the data set 6, 13, 14, 17, 24?
A. 13
B. 6
C. 24
D. 14

781. One-variable data · Easy · MC
What is the median of the data set 3, 5, 9, 10, 11?
A. 9
B. 11
C. 3
D. 7

782. One-variable data · Easy · MC
What is the median of the data set 7, 16, 17, 22, 23?
A. 7
B. 23
C. 17
D. 16

783. One-variable data · Easy · MC
What is the median of the data set 5, 6, 12, 16, 17?
A. 17
B. 12
C. 11
D. 5

784. One-variable data · Medium · MC
What is the median of the data set 4, 11, 23, 25, 26?
A. 17
B. 4
C. 23
D. 26

785. One-variable data · Medium · MC

What is the median of the data set 13, 18, 19, 21, 23?

A. 13

B. 23

C. 18

D. 19

786. One-variable data · Medium · MC

What is the median of the data set 18, 21, 22, 23, 29?

A. 22

B. 18

C. 29

D. 21

787. One-variable data · Medium · MC

What is the median of the data set 11, 13, 18, 19, 25?

A. 11

B. 17

C. 25

D. 18

788. One-variable data · Medium · MC

What is the median of the data set 11, 12, 17, 19, 22?

A. 17

B. 11

C. 22

D. 16

789. One-variable data · Medium · MC

What is the median of the data set 3, 14, 15, 18, 27?

A. 15

B. 3

C. 14

D. 27

790. Probability · Easy · MC

A bag has 6 red, 2 blue, and 4 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{1}{6}$

D. 3

791. Probability · Easy · MC

A bag has 7 red, 7 blue, and 4 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{3}$

B. $\frac{7}{18}$

C. 1

D. $\frac{25}{18}$

792. Probability · Easy · MC

A bag has 3 red, 3 blue, and 4 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{3}$

B. $\frac{13}{10}$

C. $\frac{3}{10}$

D. 1

793. Probability · Easy · MC

A bag has 5 red, 3 blue, and 1 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{5}{9}$

B. $\frac{14}{9}$

C. $\frac{5}{3}$

D. $\frac{1}{3}$

794. Probability · Easy · SPR

A bag has 7 red, 2 blue, and 4 green counters. One counter is drawn at random. What is the probability that it is red?

795. Probability · Easy · MC

A bag has 3 red, 2 blue, and 3 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{3}$

B. $\frac{3}{2}$

C. $\frac{3}{8}$

D. $\frac{1}{4}$

796. Probability · Easy · MC

A bag has 4 red, 4 blue, and 5 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{3}$

B. 1

C. $\frac{17}{13}$

D. $\frac{4}{13}$

797. Probability · Easy · MC

A bag has 8 red, 4 blue, and 5 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{3}$

B. $\frac{4}{17}$

C. $\frac{8}{17}$

D. 2

798. Probability · Medium · SPR

A bag has 7 red, 3 blue, and 2 green counters. One counter is drawn at random. What is the probability that it is red?

799. Probability · Medium · SPR

A bag has 3 red, 4 blue, and 6 green counters. One counter is drawn at random. What is the probability that it is red?

800. Probability · Medium · MC

A bag has 7 red, 7 blue, and 6 green counters. One counter is drawn at random. What is the probability that it is red?

A. 1

B. $\frac{1}{3}$

C. $\frac{27}{20}$

D. $\frac{7}{20}$

801. Probability · Medium · MC

A bag has 4 red, 2 blue, and 3 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{3}$

B. $\frac{2}{9}$

C. 2

D. $\frac{4}{9}$

802. Probability · Medium · MC

A bag has 4 red, 7 blue, and 2 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{4}{7}$

B. $\frac{4}{13}$

C. $\frac{1}{3}$

D. $\frac{7}{13}$

803. Probability · Medium · MC

A bag has 7 red, 3 blue, and 6 green counters. One counter is drawn at random. What is the probability that it is red?

A. $\frac{1}{3}$

B. $\frac{7}{3}$

C. $\frac{3}{16}$

D. $\frac{7}{16}$

804. Probability · Hard · SPR

A bag has 5 red, 5 blue, and 6 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

805. Probability · Hard · MC

A bag has 7 red, 5 blue, and 1 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

A. $\frac{7}{13}$

B. $\frac{1}{13}$

C. $\frac{7}{12}$

D. $\frac{5}{12}$

806. Probability · Hard · SPR

A bag has 9 red, 6 blue, and 6 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

807. Probability · Hard · MC

A bag has 7 red, 2 blue, and 1 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

A. $\frac{2}{9}$

B. $\frac{1}{10}$

C. $\frac{7}{10}$

D. $\frac{7}{9}$

808. Probability · Hard · MC

A bag has 7 red, 6 blue, and 6 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

A. $\frac{6}{19}$

B. $\frac{6}{13}$

C. $\frac{7}{19}$

D. $\frac{7}{13}$

809. Probability · Hard · MC

A bag has 6 red, 8 blue, and 3 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

A. $\frac{6}{17}$

B. $\frac{3}{17}$

C. $\frac{3}{7}$

D. $\frac{4}{7}$

810. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 10 joined and 12 did not. Among seniors, 8 joined and 7 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{5}{9}$

B. $\frac{4}{9}$

C. $\frac{5}{11}$

D. $\frac{10}{37}$

811. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 11 joined and 15 did not. Among seniors, 7 joined and 15 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{11}{26}$

B. $\frac{11}{18}$

C. $\frac{11}{48}$

D. $\frac{7}{18}$

812. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 18 joined and 15 did not. Among seniors, 10 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{6}{17}$

B. $\frac{9}{14}$

C. $\frac{5}{14}$

D. $\frac{6}{11}$

813. Probability · Medium · SPR

A school recorded whether students joined a club. Among juniors, 18 joined and 10 did not. Among seniors, 12 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

814. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 11 joined and 14 did not. Among seniors, 12 joined and 14 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{11}{23}$

B. $\frac{11}{51}$

C. $\frac{12}{23}$

D. $\frac{11}{25}$

815. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 13 joined and 7 did not. Among seniors, 7 joined and 11 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{7}{20}$

B. $\frac{13}{20}$

C. $\frac{13}{38}$

D. $\frac{33}{20}$

816. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 18 joined and 7 did not. Among seniors, 5 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{18}{23}$

B. $\frac{9}{19}$

C. $\frac{18}{25}$

D. $\frac{5}{23}$

817. Probability · Medium · SPR

A school recorded whether students joined a club. Among juniors, 8 joined and 15 did not. Among seniors, 6 joined and 13 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

818. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 10 joined and 13 did not. Among seniors, 10 joined and 11 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{10}{23}$

B. $\frac{3}{2}$

C. $\frac{1}{2}$

D. $\frac{5}{22}$

819. Probability · Hard · SPR

A school recorded whether students joined a club. Among juniors, 17 joined and 6 did not. Among seniors, 9 joined and 12 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

820. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 8 joined and 8 did not. Among seniors, 9 joined and 7 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{9}{17}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{8}{17}$

821. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 10 joined and 15 did not. Among seniors, 12 joined and 12 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{2}{5}$

B. $\frac{10}{49}$

C. $\frac{6}{11}$

D. $\frac{5}{11}$

822. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 12 joined and 14 did not. Among seniors, 12 joined and 9 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{3}{2}$

B. $\frac{6}{13}$

C. $\frac{1}{2}$

D. $\frac{12}{47}$

823. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 12 joined and 14 did not. Among seniors, 5 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

A. $\frac{6}{13}$

B. $\frac{5}{17}$

C. $\frac{12}{17}$

D. $\frac{4}{13}$

824. Two-variable data · Easy · MC

A line of best fit for a data set is $y = 4x + 12$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 9$?

A. 24

B. 48

C. 25

D. 12

825. Two-variable data · Easy · MC

A line of best fit for a data set is $y = -2x + 8$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 3$?

A. -14

B. 2

C. 9

D. 8

826. Two-variable data · Easy · MC

A line of best fit for a data set is $y = 3x + 7$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 10$?

A. 20

B. 7

C. 37

D. 23

827. Two-variable data · Easy · MC

A line of best fit for a data set is $y = 2x + 8$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 5$?

A. 15

B. 2

C. 8

D. 18

828. Two-variable data · Easy · MC

A line of best fit for a data set is $y = -2x + 10$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 6$?

A. 10

B. -2

C. -22

D. 14

829. Two-variable data · Easy · SPR

A line of best fit for a data set is $y = -3x + 10$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 7$?

830. Two-variable data · Medium · MC

A line of best fit for a data set is $y = 3x + 11$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 3$?

A. -2

B. 20

C. 17

D. 11

831. Two-variable data · Medium · MC

A line of best fit for a data set is $y = -2x + 5$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 10$?

A. -25

B. 13

C. -15

D. 5

832. Two-variable data · Medium · MC

A line of best fit for a data set is $y = 3x + 5$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 6$?

A. 13

B. 5

C. 14

D. 23

833. Two-variable data · Medium · MC

A line of best fit for a data set is $y = 2x + 4$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 7$?

A. 18

B. 10

C. 13

D. 4

834. Two-variable data · Medium · MC

A line of best fit for a data set is $y = 3x + 2$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 7$?

A. 23

B. 19

C. 12

D. 2

835. Two-variable data · Medium · SPR

A line of best fit for a data set is $y = 2x + 12$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 3$?

836. Inference from samples · Medium · MC

In a random sample of 600 voters, 52% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

A. Exactly 52% of all voters support the proposal.

B. The margin of error would stay 3 percentage points if the sample size were much smaller.

C. The result proves that the proposal will pass.

D. It is plausible that between 49% and 55% of all voters support the proposal.

837. Inference from samples · Medium · MC

In a random sample of 900 voters, 54% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

A. It is plausible that between 49% and 59% of all voters support the proposal.

B. Exactly 54% of all voters support the proposal.

C. The margin of error would stay 5 percentage points if the sample size were much smaller.

D. The result proves that the proposal will pass.

838. Inference from samples · Medium · MC

In a random sample of 400 voters, 57% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

A. The margin of error would stay 5 percentage points if the sample size were much smaller.

B. The result proves that the proposal will pass.

C. It is plausible that between 52% and 62% of all voters support the proposal.

D. Exactly 57% of all voters support the proposal.

839. Inference from samples · Medium · MC

In a random sample of 900 voters, 48% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

A. The result proves that the proposal will pass.

B. It is plausible that between 43% and 53% of all voters support the proposal.

C. Exactly 48% of all voters support the proposal.

D. The margin of error would stay 5 percentage points if the sample size were much smaller.

840. Inference from samples · Medium · MC

In a random sample of 1200 voters, 54% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

A. The margin of error would stay 5 percentage points if the sample size were much smaller.

B. The result proves that the proposal will pass.

C. It is plausible that between 49% and 59% of all voters support the proposal.

D. Exactly 54% of all voters support the proposal.

841. Inference from samples · Medium · MC

In a random sample of 900 voters, 61% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

A. The margin of error would stay 3 percentage points if the sample size were much smaller.

B. The result proves that the proposal will pass.

C. It is plausible that between 58% and 64% of all voters support the proposal.

D. Exactly 61% of all voters support the proposal.

842. Inference from samples · Hard · MC

In a random sample of 800 voters, 54% supported a proposal. The margin of error for this estimate is 4 percentage points. Which conclusion is best supported?

A. Exactly 54% of all voters support the proposal.

B. The margin of error would stay 4 percentage points if the sample size were much smaller.

C. The result proves that the proposal will pass.

D. It is plausible that between 50% and 58% of all voters support the proposal.

843. Inference from samples · Hard · MC

In a random sample of 400 voters, 63% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

A. It is plausible that between 60% and 66% of all voters support the proposal.

B. Exactly 63% of all voters support the proposal.

C. The margin of error would stay 3 percentage points if the sample size were much smaller.

D. The result proves that the proposal will pass.

844. Inference from samples · Hard · MC

In a random sample of 400 voters, 63% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

A. It is plausible that between 58% and 68% of all voters support the proposal.

B. Exactly 63% of all voters support the proposal.

C. The margin of error would stay 5 percentage points if the sample size were much smaller.

D. The result proves that the proposal will pass.

845. Inference from samples · Hard · MC

In a random sample of 1200 voters, 61% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

A. The margin of error would stay 3 percentage points if the sample size were much smaller.

B. The result proves that the proposal will pass.

C. It is plausible that between 58% and 64% of all voters support the proposal.

D. Exactly 61% of all voters support the proposal.

846. Evaluating statistical claims · Medium · MC

A school claims that students who join the chess club have higher math grades, so the club causes the higher grades. Why is that claim not well supported?

A. Observational results are never allowed to include numbers.

B. Students who choose the club may already differ from students who do not.

C. The school did not report the margin of error of a poll.

D. A mean cannot be computed from letter grades.

847. Evaluating statistical claims · Medium · MC

Researchers randomly select classrooms, then randomly assign those classrooms to a new reading program or the current program. The random assignment is useful mainly because it helps

A. make the margin of error equal to zero.

B. support a conclusion about the effect of the program.

C. remove the need to record any scores.

D. guarantee that every student in the country is represented.

848. Evaluating statistical claims · Medium · MC

To estimate the proportion of all students at a college who live on campus, which method is most appropriate?

A. Ask only the students in one dormitory.

B. Assign a random half of the college to move on campus.

C. Survey students at a different college.

D. Take a random sample of students from the college and record whether each lives on campus.

849. Evaluating statistical claims · Medium · MC

A poll uses a random sample of city residents. Which change would be expected to reduce the margin of error?

A. Replace the random sample with volunteers.

B. $Reporttheresultasafractioninsteadofapercent.$

C. Increase the sample size.

D. Ask the question in a different order only.

850. Evaluating statistical claims · Medium · MC

A researcher wants to know whether a new homework format causes higher quiz scores for students at one high school. Which design best supports a cause-and-effect conclusion?

A. Survey a random sample of students about which format they prefer.

B. Compare this year's quiz scores with last year's after the school changes the format for everyone.

C. Randomly assign students at the school to the new format or the current format, then compare quiz scores.

D. Ask volunteers to try the new format and compare them with students who kept the old format.

Geometry and Trigonometry

851. Angles and triangles · Easy · SPR
Two angles of a triangle measure 80° and 35°. What is the measure, in degrees, of the third angle?

852. Angles and triangles · Easy · MC
Two angles of a triangle measure 52° and 72°. What is the measure, in degrees, of the third angle?
A. 124
B. 56
C. 236
D. 90

853. Angles and triangles · Easy · SPR
Two angles of a triangle measure 78° and 52°. What is the measure, in degrees, of the third angle?

854. Angles and triangles · Easy · SPR
Two angles of a triangle measure 31° and 68°. What is the measure, in degrees, of the third angle?

855. Angles and triangles · Easy · MC
Two angles of a triangle measure 57° and 49°. What is the measure, in degrees, of the third angle?
A. 90
B. 254
C. 74
D. 106

856. Angles and triangles · Easy · MC
Two angles of a triangle measure 39° and 57°. What is the measure, in degrees, of the third angle?
A. 90
B. 84
C. 96
D. 264

857. Angles and triangles · Easy · MC
Two angles of a triangle measure 35° and 34°. What is the measure, in degrees, of the third angle?
A. 291
B. 90
C. 111
D. 69

858. Angles and triangles · Easy · MC
Two angles of a triangle measure 61° and 31°. What is the measure, in degrees, of the third angle?
A. 88
B. 90
C. 92
D. 268

859. Angles and triangles · Easy · MC

Two angles of a triangle measure 57° and 39°. What is the measure, in degrees, of the third angle?

A. 90

B. 96

C. 264

D. 84

860. Angles and triangles · Easy · SPR

Two angles of a triangle measure 71° and 44°. What is the measure, in degrees, of the third angle?

861. Angles and triangles · Easy · SPR

Two angles of a triangle measure 28° and 58°. What is the measure, in degrees, of the third angle?

862. Angles and triangles · Easy · MC

Two angles of a triangle measure 47° and 39°. What is the measure, in degrees, of the third angle?

A. 90

B. 274

C. 86

D. 94

863. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 80°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 280

B. 90

C. 100

D. 80

864. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 70°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 290

B. 110

C. 90

D. 70

865. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 36°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 324

B. 90

C. 36

D. 144

866. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 76°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 284

B. 104

C. 76

D. 90

867. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 138°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 222

B. 90

C. 42

D. 138

868. Angles and triangles · Medium · SPR

Two parallel lines are cut by a transversal. One interior angle measures 49°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

869. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 95°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 90

B. 95

C. 85

D. 265

870. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 41°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 319

B. 90

C. 41

D. 139

871. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 46°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 90

B. 314

C. 46

D. 134

872. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 37°. What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

A. 90

B. 37

C. 323

D. 143

873. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 123°. What is the measure, in degrees, of the alternate interior angle?

A. 237

B. 90

C. 123

D. 57

874. Angles and triangles · Hard · SPR

Two parallel lines are cut by a transversal. One angle measures 138°. What is the measure, in degrees, of the alternate interior angle?

875. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 96°. What is the measure, in degrees, of the alternate interior angle?

A. 84

B. 90

C. 264

D. 96

876. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 145°. What is the measure, in degrees, of the alternate interior angle?

A. 90

B. 145

C. 35

D. 215

877. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 93°. What is the measure, in degrees, of the alternate interior angle?

A. 87

B. 90

C. 93

D. 267

878. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 77°. What is the measure, in degrees, of the alternate interior angle?

A. 103

B. 283

C. 90

D. 77

879. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 20 and one leg of length 12. What is the length of the other leg?

A. 8

B. 32

C. 20

D. 16

880. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 35 and one leg of length 28. What is the length of the other leg?

A. 7

B. 63

C. 21

D. 35

881. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 61 and one leg of length 11. What is the length of the other leg?

A. 72

B. 60

C. 50

D. 61

882. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 37 and one leg of length 12. What is the length of the other leg?

A. 25

B. 35

C. 37

D. 49

883. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 20 and one leg of length 16. What is the length of the other leg?

A. 20

B. 4

C. 12

D. 36

884. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 50 and one leg of length 14. What is the length of the other leg?

A. 50

B. 48

C. 36

D. 64

885. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 25 and one leg of length 7. What is the length of the other leg?

A. 25

B. 18

C. 24

D. 32

886. Right triangles · Easy · SPR

A right triangle has a hypotenuse of length 41 and one leg of length 9. What is the length of the other leg?

887. Right triangles · Medium · MC

A right triangle has a hypotenuse of length 34 and one leg of length 30. What is the length of the other leg?

A. 64

B. 16

C. 34

D. 4

888. Right triangles · Medium · SPR

A right triangle has a hypotenuse of length 13 and one leg of length 5. What is the length of the other leg?

889. Right triangles · Medium · MC

A right triangle has a hypotenuse of length 26 and one leg of length 10. What is the length of the other leg?

A. 36

B. 26

C. 16

D. 24

890. Right triangles · Medium · MC

A right triangle has a hypotenuse of length 30 and one leg of length 24. What is the length of the other leg?

A. 6

B. 54

C. 18

D. 30

891. Right triangles · Medium · SPR

A right triangle has a hypotenuse of length 29 and one leg of length 20. What is the length of the other leg?

892. Right triangles · Hard · MC

A right triangle has legs of length 9 and 40. What is the length of the hypotenuse?

A. 49

B. 360

C. 31

D. 41

893. Right triangles · Hard · SPR

A right triangle has legs of length 24 and 7. What is the length of the hypotenuse?

894. Right triangles · Hard · MC

A right triangle has legs of length 60 and 11. What is the length of the hypotenuse?

A. 660

B. 71

C. 61

D. 49

895. Right triangles · Hard · MC

A right triangle has legs of length 11 and 60. What is the length of the hypotenuse?

A. 49

B. 71

C. 660

D. 61

896. Right triangles · Medium · SPR

An isosceles right triangle has a hypotenuse of length $4\sqrt{2}$. What is the length of each leg?

897. Right triangles · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 7. What is the length of the hypotenuse?

A. 14

B. 7

C. 49

D. 21

898. Right triangles · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 8. What is the length of the hypotenuse?

A. 24

B. 8

C. 16

D. 64

899. Right triangles · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 2. What is the length of the hypotenuse?

A. 2

B. 60

C. 4

D. 6

900. Right triangles · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 9. What is the length of the hypotenuse?

A. 81

B. 9

C. 27

D. 18

901. Right triangles · Medium · SPR

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 15. What is the length of the hypotenuse?

902. Right triangles · Medium · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 6. What is the length of the hypotenuse?

A. 36

B. 6

C. 18

D. 12

903. Right triangles · Medium · MC

An isosceles right triangle has a hypotenuse of length $8\sqrt{2}$. What is the length of each leg?

A. 16

B. 4

C. 8

D. 64

904. Right triangles · Hard · SPR

An isosceles right triangle has a hypotenuse of length $5\sqrt{2}$. What is the length of each leg?

905. Right triangles · Hard · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 10. What is the length of the hypotenuse?

A. 20

B. 100

C. 30

D. 10

906. Right triangles · Hard · MC

An isosceles right triangle has a hypotenuse of length $6\sqrt{2}$. What is the length of each leg?

A. 36

B. 6

C. 3

D. 12

907. Right triangles · Hard · MC

An isosceles right triangle has a hypotenuse of length $11\sqrt{2}$. What is the length of each leg?

A. $\frac{11}{2}$

B. 11

C. 121

D. 22

908. Right triangles · Hard · MC

In a 30°-60°-90° triangle, the side opposite the 30° angle has length 16. What is the length of the hypotenuse?

A. 256

B. 32

C. 48

D. 16

909. Right triangles · Hard · MC

An isosceles right triangle has a hypotenuse of length $15\sqrt{2}$. What is the length of each leg?

A. 15

B. 225

C. 30

D. $\frac{15}{2}$

910. Area and volume · Easy · MC

A rectangle has length 5 and width 7. What is its area?

A. 24

B. 35

C. 12

D. 37

911. Area and volume · Easy · MC

A rectangle has length 13 and width 7. What is its area?

A. 40

B. 93

C. 20

D. 91

912. Area and volume · Easy · MC

A rectangle has length 8 and width 7. What is its area?

A. 15

B. 30

C. 58

D. 56

913. Area and volume · Easy · SPR

A rectangle has length 6 and width 7. What is its area?

914. Area and volume · Easy · MC

A rectangle has length 7 and width 10. What is its area?

A. 70

B. 72

C. 34

D. 17

915. Area and volume · Easy · MC

A rectangle has length 8 and width 9. What is its area?

A. 17

B. 72

C. 34

D. 74

916. Area and volume · Easy · MC

A rectangle has length 5 and width 3. What is its area?

A. 17

B. 16

C. 8

D. 15

917. Area and volume · Easy · MC

A rectangle has length 6 and width 9. What is its area?

A. 56

B. 54

C. 30

D. 15

918. Area and volume · Easy · MC

A rectangle has length 15 and width 6. What is its area?

A. 21

B. 90

C. 42

D. 92

919. Area and volume · Easy · MC

A rectangle has length 6 and width 3. What is its area?

A. 18

B. 20

C. 9

D. 36

920. Area and volume · Medium · SPR

A triangle has a base of length 4 and a corresponding height of 4. What is the area of the triangle?

921. Area and volume · Medium · MC

A triangle has a base of length 15 and a corresponding height of 8. What is the area of the triangle?

A. 30

B. 23

C. 60

D. 120

922. Area and volume · Medium · MC

A triangle has a base of length 14 and a corresponding height of 10. What is the area of the triangle?

A. 140

B. 70

C. 35

D. 24

923. Area and volume · Medium · MC

A triangle has a base of length 7 and a corresponding height of 10. What is the area of the triangle?

A. 35

B. 70

C. 17

D. $\frac{35}{2}$

924. Area and volume · Medium · SPR

A triangle has a base of length 14 and a corresponding height of 12. What is the area of the triangle?

925. Area and volume · Medium · MC

A triangle has a base of length 16 and a corresponding height of 10. What is the area of the triangle?

A. 160

B. 26

C. 80

D. 40

926. Area and volume · Medium · MC

A triangle has a base of length 15 and a corresponding height of 6. What is the area of the triangle?

A. 21

B. 90

C. $\frac{45}{2}$

D. 45

927. Area and volume · Medium · MC

A triangle has a base of length 5 and a corresponding height of 6. What is the area of the triangle?

A. 15

B. 30

C. $\frac{15}{2}$

D. 11

928. Area and volume · Easy · MC

A rectangular prism has length 5, width 8, and height 5. What is the volume of the prism?

A. 40

B. 200

C. 18

D. 210

929. Area and volume · Easy · MC

A rectangular prism has length 3, width 8, and height 9. What is the volume of the prism?

A. 20

B. 216

C. 246

D. 24

930. Area and volume · Easy · SPR

A rectangular prism has length 4, width 6, and height 6. What is the volume of the prism?

931. Area and volume · Easy · MC

A rectangular prism has length 8, width 3, and height 9. What is the volume of the prism?

A. 20

B. 246

C. 216

D. 24

932. Area and volume · Easy · MC

A rectangular prism has length 5, width 4, and height 2. What is the volume of the prism?

A. 40

B. 11

C. 20

D. 76

933. Area and volume · Easy · MC

A rectangular prism has length 7, width 5, and height 8. What is the volume of the prism?

A. 35

B. 262

C. 280

D. 20

934. Area and volume · Easy · SPR

A rectangular prism has length 6, width 5, and height 4. What is the volume of the prism?

935. Area and volume · Easy · MC

A rectangular prism has length 3, width 3, and height 8. What is the volume of the prism?

A. 72

B. 114

C. 14

D. 9

936. Area and volume · Medium · MC

A rectangular prism has length 7, width 4, and height 5. What is the volume of the prism?

A. 28

B. 16

C. 166

D. 140

937. Area and volume · Medium · MC

A rectangular prism has length 5, width 7, and height 9. What is the volume of the prism?

A. 21

B. 35

C. 315

D. 286

938. Area and volume · Medium · MC

A rectangular prism has length 3, width 7, and height 6. What is the volume of the prism?

A. 16

B. 162

C. 126

D. 21

939. Area and volume · Medium · SPR

A rectangular prism has length 9, width 5, and height 2. What is the volume of the prism?

940. Area and volume · Medium · MC

A rectangular prism has length 8, width 5, and height 4. What is the volume of the prism?

A. 160

B. 184

C. 17

D. 40

941. Area and volume · Medium · SPR

A rectangular prism has length 9, width 7, and height 6. What is the volume of the prism?

942. Area and volume · Hard · MC

A right circular cylinder has radius 6 and height 8. The volume is $k\pi$. What is the value of k?

A. 288

B. 96

C. 48

D. 576

943. Area and volume · Hard · MC

A right circular cylinder has radius 4 and height 12. The volume is $k\pi$. What is the value of k?

A. 192

B. 48

C. 384

D. 96

944. Area and volume · Hard · SPR

A right circular cylinder has radius 3 and height 11. The volume is $k\pi$. What is the value of k?

945. Area and volume · Hard · MC

A right circular cylinder has radius 6 and height 9. The volume is $k\pi$. What is the value of k?

A. 648

B. 108

C. 324

D. 54

946. Area and volume · Hard · MC

A right circular cylinder has radius 6 and height 10. The volume is $k\pi$. What is the value of k?

A. 120

B. 720

C. 60

D. 360

947. Area and volume · Hard · SPR

A right circular cylinder has radius 5 and height 10. The volume is $k\pi$. What is the value of k?

948. Area and volume · Hard · SPR

A right circular cylinder has radius 4 and height 4. The volume is $k\pi$. What is the value of k?

949. Area and volume · Hard · MC

A right circular cylinder has radius 2 and height 12. The volume is $k\pi$. What is the value of k?

A. 16

B. 96

C. 48

D. 24

950. Circles · Easy · MC

A circle has radius 8. The circumference is $k\pi$. What is the value of k?

A. 16

B. 64

C. 8

D. 128

951. Circles · Easy · MC

A circle has radius 7. The circumference is $k\pi$. What is the value of k?

A. 7

B. 14

C. 98

D. 49

952. Circles · Easy · MC

A circle has radius 4. The circumference is $k\pi$. What is the value of k?

A. 32

B. 16

C. 8

D. 4

953. Circles · Easy · MC

A circle has radius 5. The circumference is $k\pi$. What is the value of k?

A. 10

B. 50

C. 5

D. 25

954. Circles · Easy · SPR

A circle has radius 6. The circumference is $k\pi$. What is the value of k?

955. Circles · Easy · MC

A circle has radius 12. The circumference is $k\pi$. What is the value of k?

A. 144

B. 288

C. 24

D. 12

956. Circles · Medium · SPR

What is the radius of the circle with equation $(x-2)^2+(y-7)^2=25$?

957. Circles · Medium · MC

What is the radius of the circle with equation $(x-3)^2+(y+5)^2=49$?

A. 7

B. 14

C. 3

D. 49

958. Circles · Medium · MC

What is the radius of the circle with equation $(x+2)^2+(y-5)^2=36$?

A. 6

B. -2

C. 12

D. 36

959. Circles · Medium · MC

What is the radius of the circle with equation $(x+1)^2+(y+3)^2=4$?

A. 4

B. -1

C. 2

D. -3

960. Circles · Medium · MC

What is the radius of the circle with equation $(x+3)^2+(y+6)^2=49$?

A. 14

B. 7

C. -3

D. 49

961. Circles · Medium · MC

What is the radius of the circle with equation $(x+5)^2+(y+3)^2=16$?

A. 16

B. -5

C. 4

D. 8

962. Circles · Medium · SPR

What is the radius of the circle with equation $(x+2)^2+(y-1)^2=36$?

963. Circles · Medium · MC

What is the radius of the circle with equation $(x-4)^2+(y-3)^2=49$?

A. 49

B. 4

C. 7

D. 14

964. Circles · Hard · MC

What is the radius of the circle with equation $x^2-6x+y^2+8y=-9$?

A. 3

B. 16

C. 4

D. 8

965. Circles · Hard · MC

What is the radius of the circle with equation $x^2-8x+y^2+6y=-9$?

A. 4

B. 3

C. 16

D. 8

966. Circles · Hard · MC

What is the radius of the circle with equation $x^2-8x+y^2+8y=-23$?

A. 3

B. 6

C. 4

D. 9

967. Circles · Hard · MC

What is the radius of the circle with equation $x^2-4x+y^2+8y=-11$?

A. 3

B. 2

C. 9

D. 4

968. Circles · Hard · MC

What is the radius of the circle with equation $x^2 - 6x + y^2 + 4y = 23$?

A. 6

B. 2

C. 3

D. 36

969. Circles · Hard · MC

What is the radius of the circle with equation $x^2 - 10x + y^2 + 8y = -32$?

A. 5

B. 9

C. 3

D. 4

970. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 4 and 16. Another side of the smaller triangle measures 10. What is the length of the corresponding side of the larger triangle?

A. $\frac{5}{2}$

B. 44

C. 14

D. 40

971. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 5 and 10. Another side of the smaller triangle measures 7. What is the length of the corresponding side of the larger triangle?

A. 14

B. 35

C. 19

D. 9

972. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 6 and 12. Another side of the smaller triangle measures 12. What is the length of the corresponding side of the larger triangle?

A. 24

B. 14

C. 72

D. 30

973. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 4 and 12. Another side of the smaller triangle measures 10. What is the length of the corresponding side of the larger triangle?

A. 13

B. 30

C. 40

D. 34

974. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 7 and 14. Another side of the smaller triangle measures 10. What is the length of the corresponding side of the larger triangle?

A. 70

B. 20

C. 12

D. 27

975. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 6 and 18. Another side of the smaller triangle measures 9. What is the length of the corresponding side of the larger triangle?

A. 33

B. 27

C. 12

D. 54

976. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 7 and 28. Another side of the smaller triangle measures 10. What is the length of the corresponding side of the larger triangle?

A. 70

B. 14

C. 47

D. 40

977. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 8 and 32. Another side of the smaller triangle measures 12. What is the length of the corresponding side of the larger triangle?

A. 48

B. 16

C. 56

D. 96

978. Angles and triangles · Hard · MC

Two triangles are similar. A pair of corresponding sides measure 6 and 24. Another side of the smaller triangle measures 8. What is the length of the corresponding side of the larger triangle?

A. 12

B. 38

C. 32

D. 48

979. Angles and triangles · Hard · SPR

Two triangles are similar. A pair of corresponding sides measure 5 and 15. Another side of the smaller triangle measures 7. What is the length of the corresponding side of the larger triangle?

980. Angles and triangles · Hard · MC

Two triangles are similar. A pair of corresponding sides measure 6 and 18. Another side of the smaller triangle measures 8. What is the length of the corresponding side of the larger triangle?

A. 11

B. 48

C. 24

D. 30

981. Angles and triangles · Hard · SPR

Two triangles are similar. A pair of corresponding sides measure 8 and 16. Another side of the smaller triangle measures 11. What is the length of the corresponding side of the larger triangle?

982. Angles and triangles · Hard · MC

Two triangles are similar. A pair of corresponding sides measure 4 and 12. Another side of the smaller triangle measures 9. What is the length of the corresponding side of the larger triangle?

A. 12

B. 27

C. 36

D. 31

983. Angles and triangles · Hard · SPR

Two triangles are similar. A pair of corresponding sides measure 8 and 24. Another side of the smaller triangle measures 11. What is the length of the corresponding side of the larger triangle?

984. Trigonometry · Easy · MC

In a right triangle, the side opposite angle A has length 8, the side adjacent to angle A has length 15, and the hypotenuse has length 17. What is the value of $\tan A$?

A. $\frac{17}{8}$

B. $\frac{15}{17}$

C. $\frac{15}{8}$

D. $\frac{8}{15}$

985. Trigonometry · Easy · SPR

In a right triangle, the side opposite angle A has length 7, the side adjacent to angle A has length 24, and the hypotenuse has length 25. What is the value of $\sin A$?

986. Trigonometry · Easy · MC

In a right triangle, the side opposite angle A has length 5, the side adjacent to angle A has length 12, and the hypotenuse has length 13. What is the value of $\cos A$?

A. $\frac{5}{12}$

B. $\frac{12}{13}$

C. $\frac{12}{5}$

D. $\frac{13}{5}$

987. Trigonometry · Easy · MC

In a right triangle, the side opposite angle A has length 9, the side adjacent to angle A has length 12, and the hypotenuse has length 15. What is the value of $\cos A$?

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. $\frac{5}{3}$

D. $\frac{4}{5}$

988. Trigonometry · Easy · SPR

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 35, and the hypotenuse has length 37. What is the value of $\tan A$?

989. Trigonometry · Easy · MC

In a right triangle, the side opposite angle A has length 20, the side adjacent to angle A has length 21, and the hypotenuse has length 29. What is the value of $\cos A$?

A. $\frac{21}{20}$

B. $\frac{20}{21}$

C. $\frac{21}{29}$

D. $\frac{29}{20}$

990. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 6, the side adjacent to angle A has length 8, and the hypotenuse has length 10. What is the value of $\tan A$?

A. $\frac{4}{3}$

B. $\frac{5}{3}$

C. $\frac{3}{4}$

D. $\frac{4}{5}$

991. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 35, and the hypotenuse has length 37. What is the value of $\sin A$?

A. $\frac{37}{12}$

B. $\frac{12}{35}$

C. $\frac{12}{37}$

D. $\frac{35}{37}$

992. Trigonometry · Medium · SPR

In a right triangle, the side opposite angle A has length 6, the side adjacent to angle A has length 8, and the hypotenuse has length 10. What is the value of $\cos A$?

993. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 16, and the hypotenuse has length 20. What is the value of $\cos A$?

A. $\frac{4}{5}$

B. $\frac{5}{3}$

C. $\frac{4}{3}$

D. $\frac{3}{4}$

994. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 7, the side adjacent to angle A has length 24, and the hypotenuse has length 25. What is the value of $\cos A$?

A. $\frac{7}{24}$

B. $\frac{24}{25}$

C. $\frac{25}{7}$

D. $\frac{24}{7}$

995. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 35, and the hypotenuse has length 37. What is the value of $\cos A$?

A. $\frac{35}{37}$

B. $\frac{12}{35}$

C. $\frac{35}{12}$

D. $\frac{37}{12}$

996. Trigonometry · Hard · MC

In a right triangle, the side opposite angle A has length 9, the side adjacent to angle A has length 12, and the hypotenuse has length 15. What is the value of $\sin A$?

A. $\frac{5}{3}$

B. $\frac{4}{5}$

C. $\frac{3}{5}$

D. $\frac{3}{4}$

997. Trigonometry · Hard · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 16, and the hypotenuse has length 20. What is the value of $\tan A$?

A. $\frac{4}{3}$

B. $\frac{4}{5}$

C. $\frac{3}{4}$

D. $\frac{5}{3}$

998. Trigonometry · Hard · SPR

In a right triangle, the side opposite angle A has length 5, the side adjacent to angle A has length 12, and the hypotenuse has length 13. What is the value of $\sin A$?

999. Trigonometry · Hard · MC

In a right triangle, the side opposite angle A has length 6, the side adjacent to angle A has length 8, and the hypotenuse has length 10. What is the value of $\sin A$?

A. $\frac{5}{3}$

B. $\frac{3}{5}$

C. $\frac{4}{5}$

D. $\frac{3}{4}$

1000. Trigonometry · Hard · MC

In a right triangle, the side opposite angle A has length 3, the side adjacent to angle A has length 4, and the hypotenuse has length 5. What is the value of $\sin A$?

A. $\frac{3}{5}$

B. $\frac{3}{4}$

C. $\frac{4}{5}$

D. $\frac{5}{3}$

Answer key

Student-produced responses are printed as the number to enter. Multiple-choice answers are the letter.

1. A
2. A
3. 2
4. D
5. 8
6. B
7. D
8. B
9. A
10. A
11. B
12. 6
13. B
14. C
15. 5
16. A
17. 6
18. C
19. B
20. 3
21. 10
22. B
23. 3
24. 7
25. D
26. A
27. C
28. 11
29. B
30. 3
31. A
32. A
33. C
34. D
35. B
36. A
37. A
38. C
39. D
40. D
41. D
42. 4
43. B
44. A
45. 9
46. A
47. 8
48. 4
49. 5
50. D
51. 7
52. B
53. -4
54. 7
55. D
56. B
57. B
58. A
59. -4
60. -3
61. A
62. B
63. 10
64. D
65. C
66. -1
67. A
68. B
69. D
70. A
71. B
72. D
73. A
74. A
75. B
76. B
77. B
78. C
79. D
80. A
81. C
82. C
83. -3
84. A
85. 3
86. B
87. C
88. D
89. D
90. B
91. C
92. B
93. 8
94. C
95. 8
96. C
97. A
98. B
99. D
100. A
101. C
102. C
103. A
104. D
105. C
106. D
107. A
108. D
109. C
110. A
111. A
112. B
113. D
114. A
115. A
116. B
117. B
118. D
119. A
120. A
121. A
122. C
123. D
124. C
125. B
126. C
127. A
128. C
129. A
130. D
131. D
132. B
133. A
134. B
135. A
136. A
137. A
138. D
139. D
140. B
141. B
142. A
143. A
144. B
145. A
146. B
147. D
148. B
149. B
150. C
151. A
152. A
153. B
154. B
155. A
156. D
157. C
158. D
159. A
160. C
161. C
162. C
163. D
164. B
165. 3
166. D
167. 1/2
168. A
169. A
170. D
171. C
172. 3
173. D
174. C
175. 2/3
176. B
177. C
178. 3
179. 1
180. C
181. B
182. 3
183. B
184. A
185. C
186. B
187. A
188. 16
189. C
190. A
191. C
192. D
193. D
194. C
195. D
196. 9
197. C
198. 8
199. D
200. 4
201. B
202. D
203. B
204. A
205. D
206. C
207. D
208. D
209. -1/3
210. -1/2
211. -3
212. B
213. -1/3
214. A
215. D
216. C
217. D
218. A
219. C
220. B
221. D
222. C
223. D
224. B
225. D
226. C
227. D
228. C
229. A
230. B
231. B
232. B
233. B
234. A
235. B
236. B
237. A
238. A
239. D
240. C
241. B
242. A
243. D
244. A
245. A
246. 2
247. -3
248. B
249. A
250. B
251. C
252. D
253. A
254. C
255. A

256. C
257. B
258. C
259. D
260. D
261. C
262. C
263. B
264. -1
265. B
266. D
267. B
268. 5
269. A
270. B
271. D
272. A
273. D
274. B
275. D
276. 8
277. 7
278. 7
279. D
280. A
281. C
282. B
283. C
284. 1
285. 1
286. D
287. 7
288. A
289. C
290. A
291. A
292. C
293. 9
294. A
295. 7
296. A
297. B
298. A
299. 6
300. C
301. D
302. D
303. C
304. C
305. B
306. A
307. C
308. A
309. A
310. B
311. C
312. 5
313. 15
314. 5
315. B
316. A
317. 10
318. C
319. C
320. A
321. D
322. D
323. D
324. D
325. 4
326. 7
327. C
328. B
329. B
330. D
331. 4
332. D
333. 14
334. 11
335. C
336. B
337. B
338. A
339. A
340. 13
341. D
342. B
343. A
344. B
345. 11
346. C
347. 12
348. C
349. C
350. 25
351. A
352. B
353. A
354. C
355. A
356. A
357. C
358. D
359. A
360. A
361. A
362. C
363. C
364. B
365. C
366. D
367. C
368. B
369. B
370. B
371. C
372. B
373. B
374. A
375. A
376. B
377. C
378. A
379. A
380. B
381. D
382. C
383. D
384. B
385. B
386. D
387. C
388. B
389. D
390. A
391. D
392. A
393. C
394. B
395. B
396. A
397. B
398. D
399. B
400. C
401. -3
402. 6
403. D
404. C
405. B
406. B
407. 3
408. 11
409. A
410. C
411. B
412. 7
413. C
414. A
415. C
416. D
417. C
418. 1
419. C
420. A
421. -4
422. C
423. D
424. 9
425. A
426. 9
427. D
428. C
429. C
430. A
431. A
432. 4
433. -2
434. 3
435. C
436. A
437. 8
438. 7
439. D
440. D
441. B
442. B
443. 8
444. C
445. D
446. D
447. A
448. A
449. A
450. C
451. B
452. C
453. D
454. B
455. A
456. A
457. D
458. D
459. 7
460. B
461. 9
462. D
463. B
464. -3
465. A
466. A
467. C
468. A
469. C
470. C
471. A
472. 2048
473. 64
474. C
475. A
476. 44
477. 324
478. C
479. 640
480. 63
481. B
482. C
483. D
484. B
485. 16
486. C
487. D
488. D
489. A
490. 1000
491. 800
492. D
493. 640
494. D
495. 1600
496. D
497. A
498. B
499. D
500. C
501. 18
502. C
503. 12
504. 3
505. C
506. D
507. D
508. D
509. D
510. C
511. C
512. C
513. A
514. A
515. B
516. D
517. C
518. B
519. B
520. D
521. D
522. A
523. C
524. B
525. D
526. D
527. B
528. C
529. D
530. A
531. A
532. A

533. D
534. C
535. B
536. -9
537. 19
538. A
539. 2
540. 3
541. A
542. C
543. 17
544. 0
545. D
546. 16
547. A
548. 4
549. C
550. B
551. B
552. C
553. D
554. 31
555. 42
556. B
557. C
558. D
559. A
560. A
561. C
562. B
563. D
564. A
565. A
566. A
567. C
568. C
569. C
570. B
571. A
572. B
573. A
574. 10
575. B
576. B
577. D
578. C
579. D
580. B
581. D
582. C
583. 7
584. C
585. C
586. B
587. B
588. 8
589. A
590. 5
591. B
592. A
593. D
594. A
595. A
596. B
597. 76
598. D
599. D
600. A
601. A
602. B
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639. D
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652. D
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657. C
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673. A
674. B
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806. 3/5
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808. D
809. C

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813. 3/5
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986. B
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988. 12/35
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992. 4/5
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996. C
997. C
998. 5/13
999. B
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